

Semantic Fragmentation and Stochastic Assembly, version 2

A Protocol for Decentralized Language-Model Inference over Untrusted Volunteer Nodes

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Note on affiliation and independence. This work is carried out entirely in a personal capacity as an independent researcher. The author holds academic affiliations — Pontificia Universidad Católica del Ecuador and University of Saskatchewan — unrelated to the subject matter of this work. **Neither has provided funding, materials, computing resources, personnel or institutional support for this project, and no institutional endorsement is claimed or implied.**

Relevant background. The author holds a Ph.D. in Biology (University of Saskatchewan) in avian population genomics, and works on whole-genome sequencing, variant calling and de novo assembly. The genome-assembly framing used throughout this document derives from that background; Section 3 states the criterion by which an analogy of that origin is accepted or discarded.

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Version 2.0, dated 29 September 2026. It supersedes version 1.4, dated 14 August 2026, which is kept as `WHITEPAPER_EN.md`.

The new models faced their own test. All five were measured over **473 real runs** with five local model families (§15.9). **M4 is falsified** in two independent instruments, with a controlled experiment at constant ρ . **M2 survives corrected**, and the correction withdraws a sentence from this very document. M3 survives. M1 still lacks the regime it needs and M5 is unmeasured. The document is published **with** those verdicts inside it.

Prior art. The elements E19-E24 that Section 17 declares become prior art **upon publication of this document**.

What this document does not claim is in L21 and §15.9: with a matched output budget, fragmenting costs (T09R), but the aggregate is still confounded with output length, so the abandonment criterion has **neither been met nor failed — it has not been measured**. **What it does claim, with its scope declared**, is in §15.4: extracting with the model and aggregating with code beats the model alone (T08R3), and splitting helps by itself when the task exceeds what a small node extracts reliably (T08R4).

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0. What this version is, and what changes from 1.4

Version 1.4 describes an architecture and reports what was measured of it, including two substantial withdrawals: the coherence-tax curve falling in ρ , and the reliability claim for the confidence map. This version keeps both withdrawals intact — they are not rescued through the back door — and adds what 1.4 did not have: **a theory of fragmentation itself**.

Version 1.4 took for granted that a fragment is "one of N parts of similar size" and that overlap is "a percentage". Neither was derived. This version derives them, and in deriving them changes concrete decisions in planning, packing, assembly and dispatch.

Change table.

in v1.4	in this version
The codon as the minimum unit of meaning	Withdrawn. Three of three properties fail to transfer (§5.1). The sentence is homologous to the amino acid — the alignment unit; the protein domain is homologous to the fragment — the unit of work (§5.2, §5.3)
Fragment into N parts of similar size	Replaced by cutting at weakly coupled interfaces, with L as a target and N derived: $N = \lceil \text{material}/L \rceil$ (§5.4)
Overlap as a percentage of the fragment	Replaced by F = the maximum of the longest repeat and the longest dependency crossing a boundary. Measured saving: 29% of network compute (§6.5)
ρ as a parameter to be swept	Derived from L , F and the header cost H ; and formalised as the rate of an indirect rate-distortion problem, with a second axis δ (§6.6–6.8)
Confidence map by match counting	Replaced as a mechanism by multiple alignment with learned substitution cost and consistency extension (M1, §10.5). The reliability claim remains withdrawn (§15.3, L13)
term_once enforced at the assembler	Complemented by uniqueness assignment in the plan, U-unitig style (M2, §11)

in v1.4	in this version
DAG with no control between levels	Extended with a per-level triage gate, by analogy with chaperones (M3, §8.2, §10.7)
k replicas as one parameter	Split into availability, verification and epistemic redundancy (M5, §12)
Genomic analogy as vocabulary plus one warning	Formalised as an explicit method: the instrument test and the failure-mode test (§3)
Declared elements E1-E18	E1-E18 unchanged , plus E19-E24 (§17), published with this version
The five models M1-M5, stated and unmeasured	Measured against 473 real runs over five local model families (§15.9). One is falsified (M4, in two independent instruments), one survives corrected (M2), one survives (M3), and two remain unmeasured (M1, M5)

Two hypotheses entered the literature review and **did not survive**: *mate pairs* as the homologue of global constraints (§11.1) and *fountain codes* as a redundancy mechanism (§12.1). Both are documented at greater length than their replacements, because the fall is the result.

This version arrives with a measurement campaign of its own. Version 1.4 was published with one experiment; this one is accompanied by a reference implementation (`swarmbly_ref/`), a falsification harness (`swarmbly_validation/`) and a corpus of **473 runs** over five local model families, all three published alongside the document. §15.9 reports what that campaign killed. It is worth saying here because it changes the character of the document: the claims in §15 no longer rest on a run the reader must take on trust, but on an artifact they can re-execute.

1. Introduction

1.1 The concentration is in the capital, not the knowledge

The capability to build language models is no longer scarce. Weights, training recipes and inference engines are published openly and improve monthly. What remains scarce — and what concentrates power — is the capital required to *operate* them at scale: the accelerators, the buildings, the power contracts and the interconnect.

That concentration has a measurable shape. Data centres consumed 415 TWh in 2024, roughly 1.5% of world electricity, with projections to 945 TWh by 2030 [83]. Hyperscale facilities in the United States draw from grids measured at 545 gCO₂/kWh against a 370 g national average [84]. These are the figures of an industry whose growth path runs through construction, and construction is available only to those who can finance it.

The consequence is structural rather than conspiratorial: a technology whose *knowledge* is public becomes, in practice, controllable by whoever can afford the *hardware*. Openness of weights does not democratize a capability whose operation costs hundreds of millions of dollars.

And yet the hardware already exists, distributed and idle. The flagship volunteer-computing platform aggregates approximately 700,000 active devices, 4 million CPU cores and 560,000 GPUs at an average 93 PetaFLOPS [47] — and it does so from a participant base that has shrunk from roughly a million to about two hundred thousand over two decades. That number is a *floor*, drawn from a declining niche, not a projection of what a compelling protocol could mobilise. At the level of the individual node, idle consumer GPUs serve LLM

inference at \$0.111–0.149 per million tokens on an RTX 4090, at 62–78% of H100 throughput for roughly half the cost [49].

The world's spare inference capacity is not a hypothesis. The missing piece is a protocol under which it can be used — and the reason no such protocol exists yet is a physical constraint that the next section states exactly.

1.2 The physical constraint and the reframing

Any architecture for distributed language-model inference over consumer hardware is decided, before any algorithm is chosen, by one measurement. An NVIDIA H100 SXM moves 900 GB/s per GPU over NVLink; Quantum-2 InfiniBand offers 400 Gb/s per port and 51.2 Tb/s aggregate per switch. Typical consumer upstream bandwidth is of the order of 60 Mbps. The ratio is roughly 120,000× against NVLink and 6,700× against InfiniBand. Intra-node NVLink latency is sub-microsecond and InfiniBand single-digit microseconds, against 30–170 ms of wide-area round-trip time: **four to five orders of magnitude** [22, 23, 24].

This single fact partitions the design space cleanly. Architectures that require communication *per token* are pushed against the gap on every step of generation, whereas architectures that cross it *once per unit of work* are not, and everything else in this document follows from choosing the second class.

The measured behaviour of the first class is consistent with the prediction. Petals — the reference implementation of pipeline-parallel inference over the internet — serves Llama-2-70B on three T4s at 2.29 steps/s over a 1 Gbit/s link with sub-5 ms RTT, falling to 1.57 steps/s at 100 Mbit/s and 100 ms: a 31% loss attributable to the network alone. A real geodistributed swarm of fourteen heterogeneous servers achieves 0.83 steps/s [1, 2]. Analyses of model-parallel schemes at public-internet latency find that pipeline parallelism is the best-suited model-parallel arrangement — it communicates least — and that asynchronous micro-batching does not help, because decoding is bound by KV-cache movement rather than by compute [25].

The conclusion I draw is not that pipeline parallelism was implemented poorly. It is that it is the right answer to the wrong question.

Swarmbly asks a different question: rather than *how to run one large model across many machines*, **how to run many complete small models on one large problem**.

The two are not variations of the same idea. Splitting a model creates a chain in which node k cannot begin until node $k-1$ finishes, and in which every token retraces the chain. Splitting a problem creates a set — more precisely a partial order — in which independent sub-tasks proceed concurrently and each crosses the network once. The first is bounded by $\sum_i (t_{\text{compute},i} + t_{\text{net},i})$; the second by $\max_i (t_{\text{compute},i} + t_{\text{net},i})$ plus local assembly. The structural claim is that the second is the regime in which volunteer hardware can participate at all.

The design vocabulary is borrowed deliberately from genome shotgun assembly. A request is fragmented into *reads*; each returned answer is a *contig*; adjacent contigs are joined by *overlap* and, where they disagree, by *consensus*; the plan that orders them is a *scaffold*. Section 3 states exactly what that analogy does and does not license, and does so with an explicit criterion rather than a general warning — which is the central methodological change in this version.

1.3 What this makes possible

The first measurement against the abandonment criterion is in, and **the criterion was not met**: at $\rho = 3.5$, $N = 2$, $k = 1$ over 16 held-out prompts the coherence tax is +2.30% with a

95% CI of $[-2.05\%, +7.49\%]$, and the criterion is written against the upper bound, which falls short by 2.49 points (§15.3). What survives is narrower and better founded: the **median prompt loses exactly nothing (0.00%), 11 of 16 prompts are at or below zero**, and the control at $N = 8$ fails as it was required to. The mean is manufactured by two prompts which between them are 140% of it; without them, the other fourteen average -1.06% .

Four things remain possible, and they are now motivated by a measured bimodal cost rather than by a confirmed trend.

1. Serving capacity without owning it. A participant contributes a machine that already exists and already draws power when idle. The entry requirement is a complete small model, not a shard of a large one, which places the addressable hardware pool orders of magnitude above what pipeline-parallel schemes can reach. Capacity then scales with *participation* rather than with capital expenditure.

2. A divergence map that centralization cannot produce — with its reliability claim withdrawn. Because a micro-task is answered by k nodes running *different* model families, the answers can be aligned against each other and the agreement scored per semantic unit. **A provider running one model has nothing to align.** Whether that signal carries information about correctness is a separate question, and it was answered: no, with the instrument that was used (§15.3, L13). This version proposes a different instrument (M1, §10.5) and states in advance what would kill it. Proposing a replacement does not lift the withdrawal.

3. Context bounded by the user's machine rather than by a vendor's product decision. Fragmentation relocates the context limit from a fixed window to a function of the client's assembly time and memory. With hierarchical assembly the working memory required grows logarithmically with total volume.

4. A substrate that can be audited rather than trusted. The protocol, the client, the node software and the licence are public. The share of traffic served by foundation-operated anchor nodes is published (§14.4). The energy accounting is published against a public standard (§14.3). Coherence degradation is returned with every response (§10.8). None of these is a courtesy; each is a conformance requirement.

1.4 Scope of claims

Four claims a reader might expect here are deliberately absent, and Section 16 develops each in full.

I do not claim **latency parity**: single-node speculative decoding already delivers $2\text{--}3\times$ with a proof that the output distribution is preserved [13], and no fragmentation scheme competes with that on speed. The comparison that matters is different — for a user without the hardware to run a capable model at all, the relevant axis is not *faster or slower* but *possible or impossible*.

I do not claim **unlimited context**, only a relocated and much higher limit.

I do not claim **cryptographic confidentiality**. Fragmentation is not encryption, Section 13 gives the attacks that settle it, and the protocol routes sensitive work to local execution or attested hardware.

I do not claim a **demonstrated environmental benefit**. The embodied-carbon argument is strong and the operational one is conditional; §14.3 states both.

And this version adds a fifth absence. I do not claim a **computable bound for ρ** . Section 6.7 establishes that Swarmby's problem is an indirect rate-distortion problem, from which

one inherits the *existence* of a lower bound and the qualitative monotonicity of the curve. One does not inherit a number, and saying otherwise would be exactly the kind of over-reading that Section 3 exists to prevent.

1.5 Contributions and structure

Section 2 locates the work against the prior art, including the adverse evidence. Section 3 establishes the method by which a biological homology is accepted or discarded: it is this version's methodological contribution and it organises everything that follows. Section 4 states the design principles. Section 5 develops **the semantic unit and the fragment** — how large a fragment is and where it is cut. Section 6 develops the **context budget**, now derived and formalised. Section 7 formalises the **swarm-of-small-models** thesis. Sections 8 to 10 give architecture, protocol and algorithms. Sections 11 and 12 treat cardinality constraints and the role of redundancy. Section 13 covers privacy and verification; Section 14, economics and governance. Section 15 reports what has been measured and specifies what remains, including the criterion under which I would abandon the design. Section 16 lists limitations and negative results. Section 17 declares the disclosed elements.

2. Background and related work

2.1 Decentralized inference by model partition

Petals [1, 2] distributes contiguous blocks of transformer layers across volunteers; clients hold embeddings locally and route activations through a chain of servers. I treat it as the field's pioneer rather than as a competitor: it demonstrated that peer-to-peer inference over the public internet is possible at all, which is the precondition for this work. Swarmably does not improve pipeline parallelism; it declines to use it, and that divergence is a difference of strategy rather than of quality. Hivemind and SWARM parallelism [4, 5] address fault-tolerant training over unreliable heterogeneous devices with the same underlying premise: the model is the unit of distribution.

Bittensor [6, 7] adds an incentive layer, with a consensus mechanism whose connectivity-based regularization is described as resistant to collusion of up to 50% of network weight — a formulation that, read carefully, presupposes a trust anchor.

2.2 Decentralized training over slow links

The subfield that has advanced most is training, and it advanced by attacking communication volume rather than topology. DiLoCo matches fully synchronous optimization while communicating 500× less [8]. OpenDiLoCo trained across two continents at 90–95% compute utilization [9]. INTELLECT-1 trained a 10B-parameter model on 1T tokens across up to 14 concurrent nodes on three continents with a 400× bandwidth reduction [10].

The lesson Swarmably takes is methodological: the bandwidth problem yields to compression and asynchrony.

2.3 Task-level parallelism

Skeleton-of-Thought (SoT) [12] is the direct precedent for decomposing a *prompt*: a skeleton prompt produces a list of points, each expanded independently and in parallel. It reports up to 2.39× speedup, and — this matters more — it reports its own damage: quality improves on knowledge, generic, common-sense, roleplay and counterfactual questions, and degrades on

maths, coding, writing and Fermi estimation. The authors state the structural cause without hedging: "SoT currently ignores the dependencies between points."

Their response was not to defend the method but to gate it. SoT-R [12] adds a router that decides per question whether to decompose at all; a trained 120M RoBERTa router suffices, and it is trained with a Tversky loss precisely to penalize false positives.

Descendants refine the idea. APAR [16] has the model plan its own parallel branches. PASTA [17] learns an annotation language for semantically independent spans and reports geometric-mean speedups of 1.21–1.93× at a length-controlled win-rate delta of +2.2% to −7.1%. Plato/ASGD [18] replaces the flat list with a **dependency graph** and reports a 68% throughput gain. Hogwild! Inference [19] takes the opposite tack: concurrent workers sharing a live KV cache.

ParallelBench [20] supplies the theory: the conditional-independence assumption underlying parallel generation "inevitably degrad[es] generation quality when dependencies are strong." Tran and Kiela [21] give the information-theoretic version via the data processing inequality.

What is missing from that literature is my subject: nobody has combined prompt-level decomposition with dispatch to untrusted volunteer nodes. That intersection, and not either half of it, is what this document discloses.

2.4 The nearest neighbour, and the adverse evidence

Two things have changed since v1.4 and both must be faced head-on.

The nearest neighbour now exists. SWARM-LLM (Dahshan, Mamun & Debnath, VTC 2026) [134] routes a query to a local SLM, to several peer SLMs at the edge with **weighted consensus where lower-uncertainty nodes weigh more**, or escalates to the cloud. It reports ~28% cloud usage and accuracy on hard questions rising from 0% to 15% over a load of **50 queries**. That is weak empirical evidence — 50 queries — but it claims the territory, and it must be cited and differentiated from. The difference is that its consensus is a scalar weighting per node, and Swarmby's is a position-to-position correspondence (§10.5): theirs says *how much* to trust each node, ours says *where* they diverged.

Semantic agreement across different models is already used as a signal. Soiffer, Kolawole & Smith (2025) [101] use agreement among smaller models as a deferral signal toward a larger one: "when independently generated outputs are semantically consistent — even if lexically distinct — their agreement suggests the underlying meaning is reliable". Architecturally it is the closest thing to the confidence map. It also clusters rather than aligns, which is the opening §10.5 occupies.

And the serious criticism of the whole family. "Why Do Multi-Agent LLM Systems Fail?" (Cemri et al., 2025) [90] builds the MAST taxonomy: **14 failure modes in 3 categories** — system design issues, inter-agent misalignment, task verification — from more than 1,600 annotated traces over **7 frameworks**, with 150 traces validated by expert annotators at **κ = 0.88**. "If Multi-Agent Debate is the Answer, What is the Question?" (Zhang et al., 2025) [135] finds that multi-agent debate methods "fail to consistently outperform single-agent baselines such as Chain-of-Thought and Self-Consistency, even when consuming additional inference-time compute", across 9 benchmarks, 4 models and 5 methods.

Swarmby's honest answer is not that its architecture is immune. It is that **MAST's three categories are exactly the three this version addresses**:

MAST category	where it is addressed here
System design issues	§5.4 interface cutting rule; §11.3 uniqueness assigned in the plan

MAST category	where it is addressed here
Inter-agent misalignment	§10.5 alignment with semantic substitution; §9.2 global contract
Task verification	§10.7 per-level triage gate

That a taxonomy built independently over 1,600 traces coincides with the three areas the biological homologies pointed at is, if not a validation, at least a convergence worth declaring.

Zhang et al.'s criticism is moreover more specific than it appears: it targets **debate** — agents arguing to converge — and not **decomposition** — agents solving disjoint parts. Swarmblly does not debate. But the lesson transfers all the same, and this version adopts it as a requirement: **additional compute must be justified against a single-agent baseline with self-consistency**, not against the naive monolithic model. Section 15.6 folds it into the evaluation protocol.

2.5 Prompt compression as a rate-distortion problem

Nagle et al. (NeurIPS 2024) [123] formalise **prompt compression** as a rate-distortion problem for black-box language models. Their rate is $E[\text{len}(M)/\text{len}(X)]$ — expected compressed prompt length over expected original length — which is **structurally the same quantity as ρ** . Their distortion is performance degradation under logarithmic or 0/1 loss. They derive the distortion-rate function via the dual of a linear program and report a large gap between current methods and the optimal strategy.

The consequence for this document is direct and limiting: **the contribution cannot be " ρ is a rate-distortion problem", because that is published**. The contribution has to be ρ **under fragmentation with distributed reassembly**, which is a different and harder problem, and one Nagle et al. do not cover. Section 6.7 develops it that way, and 6.8 adds the second axis their framing does not need and ours does.

2.6 Genome assembly

Lander-Waterman coverage statistics [26] give, for a genome of length G sampled by N clones of length L with minimum detectable overlap fraction θ :

$c = L \cdot N / G$		(coverage redundancy)
$P(\text{base uncovered})$	$= e^{(-c)}$	
$E[\# \text{ apparent islands}]$	$= N \cdot e^{(-c \cdot \theta)}$	
$E[\# \text{ clones per island}]$	$= e^{(c \cdot \theta)}$	

This model transfers, but only after one correction that earlier versions of this work got wrong. The relevant difference between genome assembly and text assembly is **not** that the target sequence is *known*: in *de novo* assembly no reference exists. The real difference is narrower and more useful: in genomics there exists **a single physical molecule of which every read is a sample**. That uniqueness is what guarantees that two true overlaps are reconcilable. In free text generation there is no such guaranteeing object.

But the guaranteeing object can be manufactured. If the plan $\mathfrak{p}$ and the global contract $\mathfrak{r}$ are fixed *before* any generation occurs and are treated as the reference, then a common underlying object exists again and coverage statistics become applicable. Section 7.4 develops this, and it is the point at which the analogy stops being a naming convention and becomes a derivation.

There remains one genuine transfer from the assembly literature, and it is a warning. In *de Bruijn* assembly a repeat longer than k collapses into a single graph node; the Eulerian path stops being unique and the number of valid reconstructions grows combinatorially with repeat

count [29]. Twenty years of practice established the consequence: **repeats, not coverage, are the binding constraint** [30, 31]. Section 6.4 turns that warning into an inequality with two forms, and into a measured datum from the project itself.

3. The method: what makes a homology useful

This section is new and is version 2's methodological contribution. Without it, Sections 5, 6, 10, 11 and 12 would be a catalogue of attractive analogies.

3.1 The problem with analogies

Swarmbly was born of an analogy: a request too large for a small model resembles a genome too long for a sequencer. It is broken up, read in parts, reassembled.

Analogies of that kind are productive at first and dangerous afterwards. They are productive because they import mature vocabulary for a new problem. They are dangerous because the vocabulary arrives with connotations that do not transfer, and because an analogy that sounds good resists scrutiny precisely for that reason.

Version 1.4 stated the caution — no genome assembly algorithm runs inside Swarmbly — but offered no criterion for deciding when a transfer is real. This section proposes one, and the rest of the document applies it case by case.

3.2 The instrument test

A homology is useful when it brings an instrument: a procedure, an inequality or a number that can be applied to the new problem. It is not useful when all it brings is a way of talking.

The clear case is the coverage equation. Lander & Waterman (1988) derive that the expected number of islands is $N \cdot e^{(-c\theta)}$ with $c = \ln/G$. That derivation hands over a number — how many replicas are needed — where before there was a guess. It is an instrument, and Section 7.4 uses it as one.

The opposite case, and it has to be said because it comes from the project itself: calling a fragment of an answer a *contig* does nothing. It is vocabulary. It is kept because it is convenient, not because anything is derived from it.

3.3 The failure-mode test

Applying the instrument test to the homologies this project uses, a pattern emerged that had not been looked for:

Homologies that transfer come accompanied by their failure mode. Those that do not transfer bring a mechanism without its associated pathology.

The minimum-overlap criterion comes with *mis-assembly*: if the overlap does not exceed the repeat, the assembler glues the wrong stretches together. The protein domain comes with its interface condition: independence is lost when the interface is large and tightly packed. Chaperone triage comes with degradation: what cannot be recovered is destroyed, not patched in.

By contrast the codon — which §5.1 discards — was proposed as "the minimum unit of meaning" with no statement whatsoever about what happens when it is violated. Fountain

codes — §12.1 — were proposed for their recovery property with no condition on when that property ceases to exist.

The reason is that a mature field does not discover an isolated mechanism: it discovers a mechanism **and** the cases where it fails, and it usually publishes the second with more care than the first. A transfer that brings only the good part is importing the conclusion without the work.

3.4 The operational test

question	if the answer is no
Does it bring a procedure, an inequality or a number?	It is vocabulary. Use it as vocabulary and derive nothing from it.
Does it bring a statement of what happens when the condition is violated?	It is half-imported. Find the failure mode before building on it.
Do the source field's conditions hold here?	Invalid transfer. Say why, which is usually informative.

The five homologies this document uses pass all three questions. The two that are discarded fail the second, the third, or both — and in both cases the diagnosis of *why* they fail turned out to be more useful than the homology would have been.

4. Design principles

P1 — Cross the network once per unit of work. The only defensible performance argument available to a volunteer network.

P2 — The orchestrator may refuse. A system that fragments every request is strictly worse than SoT was in 2023, because SoT shipped a router. Fragmentation is a decision with an asymmetric cost function, not a default.

P3 — Model dependencies explicitly. Plans are directed acyclic graphs. Parallelism is the width of a level, not the size of the task set.

P4 — Select before you synthesize. Where several candidate fragments exist, choose one and splice it. Rewrite only where a seam actually fails.

P5 — Calibrate every threshold. No fixed cosine cutoffs, no assumed redundancy ratios. Thresholds are derived from labelled data per model and per domain, with asymmetric objectives.

P6 — Report the tax. Every assembly returns a coherence audit. A protocol that hides its own degradation cannot be evaluated.

P7 — Verify cheaply or not at all. Verification that costs a significant fraction of inference destroys the economics.

P8 — Route by sensitivity, do not pretend to encrypt. Confidentiality is a routing decision with three lanes, not a property claimed for fragmentation.

This version adds three, and all three come out of the new sections.

P9 — Cut where the coupling is weak, not where the count falls. A fragment boundary is a decision about the interface, not about size. $\perp$ is a target; the interface is the rule. (§5.4)

P10 — Resolve global constraints in the representation, not in the output. A constraint no node can satisfy in isolation is assigned in the plan, where it can be satisfied by

construction. (§11.3)

P11 — One parameter, one purpose. *k* today serves three distinct ends — availability, verification and epistemic redundancy — and pays for all three at the price of the most expensive. They are separated. (§12.4)

5. The semantic unit and the fragment

This section does not exist in v1.4. It answers how large a fragment is and where it is cut, and the planner (§10.2), the context budget (§6) and the assembler (§10.4) all depend on it.

5.1 The codon is not the homologue

The original proposal took the **codon** — three nucleotides translated into one amino acid — as the homologue of the minimum unit of meaning in text, and derived from it that there exists a minimum fragment size below which meaning is destroyed.

The conclusion is correct. The derivation is not. The codon has three structural properties, and the sentence has none of them:

property of the codon	the sentence
Fixed width — exactly 3 nucleotides	Variable. Measured over the project corpus: p25 = 9 tokens, p75 = 25. Nearly 3× range.
Reading frame — non-overlapping; a shift ruins everything downstream	There is no frame. A displaced cut does not destroy the rest of the text.
Deterministic map — GGA is always glycine	The same idea admits infinitely many textual surfaces.

Three out of three. The transfer is not partial: it is null on all three properties that define the codon. Applying §3.4, it also fails the second question — it never came with a statement of what happens when it is violated.

5.2 The amino acid is the homologue of the alignment unit

The third row points at where it does fit. The amino acid is a unit of **variable width, self-delimiting**, with **properties of its own**, and it is **the unit of the folded structure**. The sentence shares all four.

And the distinction is not terminological, because computational biology has **two alignment technologies** and the choice of homologue determines which one is inherited:

- **Nucleotide alignment** — identity or not. A against G is a disagreement, full stop.
- **Protein alignment** — substitution matrices. Henikoff & Henikoff (1992) [92] build BLOSUM by counting aligned amino acid pairs in ungapped blocks and computing, for each pair, $s_{ij} = \log_2(q_{ij} / e_{ij})$, the log ratio of observed to expected frequency, in half-bit units.

That formula is the instrument. It says that the interchangeability of two units is not postulated: **it is counted**, over cases already known to be homologous.

For Swarmbly the consequence is direct. Two nodes that answer

"The tide window closes at noon." "The channel shuts at midday."

are not in disagreement. Under nucleotide-style comparison they are a total disagreement; under a substitution matrix they are a **conservative substitution**. The v1.4 assembler —

which compares by cosine similarity against a threshold — is operating in the nucleotide regime on material that is protein.

5.3 The domain is the homologue of the fragment

This is the most important correction in this version.

The amino acid is not the homologue of the **fragment**. No amino acid folds alone or has function alone. The unit that does is the **domain**.

Porter & Rose (2012) [110] give the rigorous definition: a domain is "a contiguous segment of the folded protein whose m-value remains largely unchanged when that segment is excised from its parent structure", and they equate domain with "cooperative folding unit; i.e., its cooperativity depends primarily on intra-segment, not inter-segment, interactions".

Read the second clause carefully, because **it is the specification of a good task fragment**: a piece whose resolution depends on what it contains and not on what its neighbours contain.

The reference work on multidomain proteins (Han et al., 2007) [111] confirms that more than 70% of eukaryotic proteins are multidomain and that domain folding is independent — **but with an explicit condition**: "where the interface is small and poorly packed, or unstructured, folding of the domains is independent".

5.4 The interface condition is the cutting rule

That condition is not a footnote caveat. It is the instrument, and it is P9:

Fragment where the coupling between fragments is weak. A fragment boundary is a decision about the interface, not about size.

This reorders the design. The planner must not cut every L sentences and hope it works out: it must **look for the points of weak coupling** and cut there, with L as a target and not as a rule. A cut that splits a strong dependency produces two fragments that are not domains, and the independence guarantee applies to neither.

The project already has a recorded incident that is exactly this failure: the segmenter separated a question from the material that answered it. That was not an implementation error, it was a cut across a strong interface. Under the v1.4 formulation — cut into N parts of similar size — that failure is structural and recurrent. Under the interface rule it is detectable before dispatch, because the planner can measure the coupling it is about to cut.

And there is a datum from the project pointing the same way. In the measurement of §15.3, the only two prompts where $N = 2$ is *worse* than $N = 8$ are the two that manufacture the mean. More fragments *helping* is not what a coherence-tax story predicts; it is what a **partition-quality** failure predicts — a two-way split putting a seam somewhere costly, which cutting eight ways happens to avoid. Under P9 that stops being luck.

5.5 Functional autonomy is quantified, and it is not total

Bashton & Chothia (2007) [112] compare homologous domains that appear both in single-domain and in multidomain proteins, over 70 unique domain pairs in 45 protein sets: approximately **three quarters preserve their function** when the context changes; **a little under one sixth changes substantially**.

The number bounds the expectation. "A fragment solved in isolation is worth the same as in context" is true most of the time and false one time in six or seven. A design that assumes

total autonomy is assuming something nature does not deliver even in the system the idea was borrowed from.

5.6 Size: there is a band, there is no number

Is there a natural domain size that would suggest a natural L for text? The honest answer is that **a characteristic band exists and a number does not**, and the reason it does not is instructive.

Zhang et al. (2005) [113] map sequence-defined domains against structure-defined domains, over the same proteins: **Pfam averages 96 residues; SCOP averages 174**. Almost double, over the same material, in the same paper. The difference is not noise: Pfam cuts by sequence family and SCOP by structure. **The definition determines the number**.

Schaeffer et al. (2023) [114], classifying domains over predicted structures, report **99.8 ± 64.8 residues** — a standard deviation that is 65% of the mean.

The floor, by contrast, is sharp. Porter & Rose set **25 residues** as the minimum, "approximating the size of a supersecondary structure unit"; UniDoc (Zhu et al., 2023) [115] uses 30 as a parsing constraint.

Both lessons transfer cleanly:

1. **The floor is hard and principled.** Below a certain size nothing folds cooperatively. For text, this predicts that an $L_{\min}$ exists below which a fragment is not independently solvable, and that this floor is sharper than the optimum. Section 6.5 gives an independent reason for the same floor, from the header-cost side.
2. **The optimum is a wide, definition-dependent band.** Any claim that $L^* = 50$ is a number is an over-reading. What can be expected is a band spanning a factor of 3 to 6, and the definition of "unit" chosen will move its centre.

5.7 The fundamentals of the semantic unit

1. The minimum alignment unit is the **sentence**, homologous to the amino acid: variable width, self-delimiting, with meaning of its own, and the unit over which substitution is computed.
2. The fragmentation unit is the **task domain**, homologous to the protein domain: the segment whose resolution depends on internal interactions and not on its neighbours.
3. The fragment boundary is chosen **where coupling is weak**, with L as a target in sentences and not as a division rule.
4. L is a property of the **node class**, measured and revisable. N is derived: $N = \lceil \text{material} / L \rceil$. This inverts the v1.4 formulation, where N was the parameter and size the consequence.
5. There is a hard $L_{\min}$; the optimum is a band.

6. The context budget

This is the project's central constraint. Version 1.4 stated it and swept it; this version derives it and formalises it.

6.1 Definition

Let a request P be decomposed into micro-tasks $T = \{t_1 \dots t_N\}$ with dependency DAG $D = (V, E)$. Each dispatched packet is

$$K_i = (\Gamma , \sigma(R_j : (t_j \rightarrow t_i) \in E) , t_i)$$

where Γ is the *global contract* — objective, audience, register, output format, target length, forbidden vocabulary, session identifier — and $\sigma(\cdot)$ summarizes the results of t_i 's predecessors.

Define the **context budget**

$$S = |\Gamma| + E[|\sigma(\cdot)|] \quad (\text{tokens of shared context per packet})$$

and the **contextual redundancy ratio**

$$\rho = (\sum_i |K_i|) / |P|$$

ρ is what the operator pays. In v1.4, S was what the operator chose. In this version it is not: §6.5 derives ρ from L , F and the header cost, and what the operator chooses are those three.

6.2 The four-way tension

Four desiderata are functions of S , and they do not agree:

Desideratum	Behaviour in S	Mechanism
Assembly coherence	increases	Workers share the decisions that make fragments compatible. Absent a contract, one worker renders the scene in one register and another in a second [33]
Fragment verifiability	increases	A verifier cannot judge whether a fragment is faithful to a specification it was not given
Privacy by decontextualization	decreases	Γ is the session's objective, audience and constraints. A node holding Γ holds the shape of the request
Required worker capability	plausibly decreases	Context supplied in-prompt substitutes for knowledge held in parameters — stated as a hypothesis in §7.3, not as a result

And ρ , hence cost, grows approximately linearly in $N \cdot S$.

6.3 The falsifiable core

Proposition (Context Budget). Swarmbly is viable if and only if there exists a context budget S^* that simultaneously satisfies: a coherence tax below the application's tolerance; a leakage bound below the user's tolerance for the applicable sensitivity lane; a verification accuracy above the protocol's security requirement; and a worker-capability requirement met by commodity small models — all at a ρ whose cost remains below the value of the aggregated capacity.

This is the whole project stated as one testable claim, and it is why the reference implementation measures a curve rather than demonstrating a system.

It also predicts something useful. Because S is shared across all four, **any improvement that raises coherence per token of context is worth more than an improvement that raises coherence per token of output** — it buys progress on privacy and cost simultaneously. This makes contract compression, and not fragment quality, the highest-leverage research direction in the design.

6.4 Flanks, repeats and the information floor

Version 1.4 treated overlap as a percentage inherited by analogy. It is not, and the information-theoretic literature says exactly what it is.

Overlap solves a different problem here. In *de novo* assembly, overlap exists above all to **discover order**: nobody knows which part of the genome each read came from. Swarmably does not have that problem — the orchestrator creates the fragments and knows their order. Two reasons remain, and only one is genomic: **transition continuity** (that the end of one fragment meets the beginning of the next) and **repeat detection**.

The minimum-overlap criterion has two forms. The folk statement — "overlap must exceed the longest repeat" — is correct but weak. Bresler, Bresler & Tse (2013) [102] sharpen it into two distinct conditions:

Sufficient condition for a greedy algorithm:

$$L > \ell_{\text{repeat}} + 1$$

"GREEDY reconstructs the original sequence if every repeat is bridged." That is a statement about **one particular algorithm**.

Necessary, information-theoretic condition:

$$L > \max\{\ell_{\text{interleaved}}, \ell_{\text{triple}}\} + 1$$

where an *interleaved* repeat is a pair of repeats whose positions interleave, and a *triple* repeat is a subsequence that appears three times. Below that threshold **no algorithm** can reconstruct, because two different sequences produce identical read sets.

The distinction matters more than it appears. The first form says "do better". The second says **there exists a class of inputs where the fragments simply do not contain the information needed to reassemble correctly, however clever the assembler is**.

Motahari, Bresler & Tse (2013) [103] formalise the threshold: for i.i.d. sequences there is a sharp transition according to whether the normalised read length exceeds the order-2 Rényi entropy of the source. Above it, "the obvious coverage condition is also sufficient for reconstruction"; below it, no amount of coverage suffices.

The parallel this enables, and which this version adopts: coverage — how many replicas — and solvability — whether the fragments contain the information — are **two distinct questions**, and the first does not imply the second. Version 1.4 reasoned about k and kept silent about solvability. The coverage model of §7.4 is still correct and is still only half the matter.

A repeat is a property of a pair, and the project has already measured it. The semantic homologue of a genomic repeat is a recurring phrase or template. Two fragments that cannot see each other repeat the same structure, and the assembler cannot detect it because **a repeat is a property of a pair, not of a fragment**. Bresler et al.'s formal definition is inherently relational: a repeat "is a subsequence that appears twice" at positions t_1 and t_2 . No individual read can be inspected to determine whether it lies in a repeat.

And the project **has already measured the same phenomenon**. The `no_repeated_ngram` constraint is the irreducible class in its composition measurements: **4 of 12 against 11 of 12 for the monolithic arm**, and no context-allocation policy reaches it. It was the defect that would not yield. The reason, now nameable, is that no allocation *can* reach it: a fragment cannot avoid repeating what it cannot see. Section 11 gives the mechanism that can.

6.5 The flank F and the derivation of ρ

From the above comes the operational criterion:

F is the maximum of two quantities measured over the corpus: the longest repeatable unit the assembler must be able to detect, and the longest dependency that crosses a fragment boundary.

Reference estimate for the project corpus: **$F \approx 5\text{--}10$ sentences.**

With F defined by measurement and L as a property of the node class, the context budget is **derived** rather than swept:

$$\rho \approx (L + 2F) / L + H / (L \cdot s)$$

where H is the fixed per-packet cost — contract, glossary, formatting instructions — measured at **≈ 39 tokens** over the project corpus, and $s \approx 15$ tokens per sentence.

With $L = 50$, $F = 10$: **$\rho \approx 1.45$** . With a flank inherited by analogy, $F = 25$: **$\rho \approx 2.05$** .

29% of the compute of the whole network, from the single decision to measure the flank instead of copying a percentage. And since §6.4 says exactly what to measure, the saving relaxes no guarantee: F is set by the observed repeat and dependency, not by convenience.

The second term explains something the project observed without being able to name it. The header is paid **in full per packet**, so its relative weight is inversely proportional to L . With 35-token fragments — the actual fragment size in the project's composition measurements — the 39-token header **weighs more than the material**. That is the regime in which measurements were taken for months, and it explains why ρ appeared to have a high floor: it was not a property of the protocol, it was a property of fragmenting too finely.

This has a direct consequence for v1.4's metric. That document set "operating $\rho < 2.0$ " as a target and then reported it was not attainable on its corpus, because the packing floor at $N = 2$ was 1.42–1.68. With the derivation above, that floor stops being a brute fact and becomes a prediction: it was the term $H/(L \cdot s)$ dominating because L was too small. The correct target is not an absolute ρ but **the distance to $(L+2F)/L$** , which is what can be reduced without losing guarantees.

6.6 Why formalise ρ

ρ has functioned as an empirical quantity: it is measured, compared, reported. The derivation above is already an advance over treating it as a dial. But deriving it does not say **whether a floor exists**, nor what it depends on. Rate-distortion theory is the framework in which that question has an answer.

Background concept. Rate-distortion theory (Shannon, 1959) [121] answers: given that I am going to transmit an imperfect version of something, what is the minimum information rate needed for the imperfection not to exceed a given level?

Shannon defines a distortion matrix in which " d_{ij} measures the 'cost' or 'distortion' if letter i is reproduced at the receiver as letter j ", and $R(d^*)$ as the minimum rate subject to the average distortion not exceeding d^* . Cover & Thomas [122] write it as

$$R(D) = \min I(X; \hat{X})$$

minimised over conditional distributions $p(\hat{x}|x)$ satisfying the expected distortion constraint. Their Theorem 13.2.1 establishes that the rate-distortion function of an i.i.d. source with bounded distortion **equals** the associated informational function.

The shape of the curve is the essential part: $R(D)$ decreases with D . Less fidelity demanded, less information needed. And there is a floor: below $R(D)$ the distortion D is unreachable, whatever one does.

6.7 ρ as the rate of an indirect problem

The natural objection is that classical rate-distortion assumes a source that is reconstructed, and Swarmbly does not reconstruct the request: it produces an answer.

The theory has a name for that situation: the **indirect** or **remote** rate-distortion problem, in which "the encoder cannot observe the source directly but obtains only noisy observations" — a formulation that goes back to Wolf & Ziv.

The mapping is exact, and it is this section's formal contribution:

element of the theory	in Swarmbly
Source Y^*	The ideal answer to the request. Never observed , neither by the orchestrator nor by any node.
Observation X	The request P . It is what the encoder <i>does</i> see: a description of Y^* , not Y^* .
Encoder	The fragmentation policy: router, planner, packer. It produces $\{K_1 \dots K_N\}$.
Channel	The node network. Each node applies a stochastic transformation $K_i \rightarrow R_i$.
Decoder	The local assembler, which produces $\hat{Y}$ from $\{R_i\}$.
Rate	$\rho = \Sigma$
Distortion	$d(Y^*, \hat{Y})$ — task loss, not reconstruction fidelity. It is Nagle et al.'s move, and it is the right one.

With this, the defensible claim can be stated:

M4 — ρ is the rate of an indirect rate-distortion problem whose distortion is task loss. From this one inherits the existence of a lower bound and the qualitative monotonicity of the curve. One does not inherit a computable bound.

What cannot be claimed, and why. Three limits, stated before a reviewer states them.

The bound exists but is not computable here. Nagle et al. obtain a number because they restrict to token erasure over a known prompt with a black-box LLM, and solve a finite linear program. Swarmbly's encoder is a decomposition policy and its decoder is a language model; there is no tractable $p(x)$ over the task space and no single-letter distortion. One may claim the bound **exists**; not that it has been computed.

The theorem is asymptotic in i.i.d. blocks. A user request is one realisation, not a long sequence. The converse gives no per-request guarantee.

And the most important one: distortion is not caused by rate alone.

6.8 The second axis: dependency density

This is the section's own contribution, and it comes out of a datum from the prior art.

Skeleton-of-Thought [12] generates a skeleton and expands the points in parallel. It achieves speedups of up to **2.39×**. And its quality degradation **is not uniform**: it improves "diversity and relevance while hurting immersion and coherence", and **fails on maths and coding**.

Those are precisely the tasks where sub-answers are interdependent. The degradation does not track rate: it tracks **how much was cut across dependencies**.

A curve $D(\rho)$ indexed only by ρ cannot capture that. At least a second axis is needed:

$$D = D(\rho, \delta)$$

where **δ is the dependency density of the decomposition**: some measure of how many necessary relations cross a fragment boundary, normalised by the number of fragments.

This closes the document's argument from three sides at once. **The weak-interface cutting rule (P9, §5.4) is, in this language, the minimisation of δ** : cutting where coupling is weak is exactly choosing the decomposition of lowest dependency density for a given ρ . And the Bresler et al. information floor (§6.4) is the statement, in assembly language, that **there exists a region of the (ρ, δ) plane where no acceptable distortion is reachable**, because the fragments do not contain the information.

The three formulations — weak interface, information floor, dependency density — are the same claim in three vocabularies. That they converge from three independent literatures is the reason to believe it.

And it is falsifiable without further theory. Fixing δ and varying only ρ (same cut, different flank) should produce a curve monotonically decreasing in distortion. Fixing ρ and varying δ (same budget, cuts of different coupling) should produce variation in distortion **at constant rate** — which is the result that would kill the idea that ρ suffices as an axis. And the unreachability region should manifest as a distortion floor that no ρ improves.

6.9 Why the earlier formulation was inadequate

An earlier version of this design specified a fixed redundancy target ($c_{\text{sem}} > 1.2$, "20% intentional redundancy") derived by analogy from Lander-Waterman, and a fixed seam threshold ($\tau_{\text{sem}} = 0.85$) on embedding cosine similarity.

Both are withdrawn. The first for the reasons in §2.6 and because the step from a ratio to a redundancy percentage holds only if all excess length is flank, which ceases to be true the moment a global contract is introduced. The second because contextual embedding space is anisotropic — randomly chosen words already exhibit high mean cosine similarity [34] — because cosine similarity in regularized models can be "arbitrary and therefore meaningless" [35], because no embedding model dominates across task types [36], and because the reference library for this operation deliberately recommends **no threshold at all** and warns that the similarity is asymmetric [37].

This version adds a third reason, and it is the deepest: **the cosine threshold is a nucleotide-type instrument applied to protein-type material** (§5.2). It is not that the threshold is badly calibrated; it is that the correct operation is a learned substitution matrix, and §10.5 specifies it.

7. Swarm intelligence with small models

7.1 The thesis

Swarmbly dispenses with the monolithic model entirely. No participant holds a shard of a 70B or 400B network. Worker nodes run *complete, independent* small language models — typically 1–8B parameters, quantized, on consumer GPUs, unified-memory laptops or multi-core CPUs

— and each receives a decontextualized micro-task and answers it in isolation. The client runs a small model too, but its competence is a different one: not world knowledge, but *logic and syntax*, in the sense of understanding the request, planning its decomposition and suturing the returned fragments.

The claim is that advanced capability need not reside in a single large network, but can be the arithmetic result of coordinating many small ones — the answer emerging, as a genome does, only at assembly.

This is a strong claim. It is also partly supported, partly unsupported, and partly false as usually stated. This section separates the three.

7.2 Coverage and conversion

I propose decomposing swarm performance into two independent factors.

Coverage c is the probability that *at least one* worker response to a given micro-task is acceptable, and **conversion** v is the probability that the orchestrator, given that acceptable fragments exist, selects and assembles them into an acceptable whole.

For a plan of N micro-tasks with per-task coverage c_i and a global conversion factor v :

$$Q_{\text{system}} \approx v \cdot \prod_i c_i$$

The product over i is the uncomfortable term — it is why N cannot grow freely — but the factor that decides the architecture is v .

Coverage scales with the swarm. Conversion does not.

The evidence for the first half is strong. Repeated sampling raises coverage log-linearly over four orders of magnitude of sample count: on SWE-bench Lite with DeepSeek-Coder-V2-Instruct, 15.9% at one sample rises to 56% at 250 samples, beating a 43% single-sample state of the art [38].

The evidence for the second half is equally strong and is usually overlooked. The same work states that "majority voting and reward models plateau beyond several hundred samples" — coverage keeps rising and the *ability to cash it in* saturates [38]. Judge-based selection over diverse teams achieves an 81% win rate against a single-model baseline, while homogeneous teams achieve 51.2% — chance — and produce 100% ties across 756 verdicts [39]. And in the study closest to Swarmby's own architecture, an 8B multi-agent system ties a 32B single agent with tools on GAIA (23.0 vs 23.0) and beats it on AIME (55.0 vs 45.0), running 4.2× faster — but performance is "primarily driven by orchestrator capacity rather than sub-agent capacity" [40].

7.3 Three consequences, and an unsupported hypothesis

(a) The client is the ceiling, not the network. The marginal value of the $(N+1)$ -th node is bounded above by the orchestrator's ability to select among what already arrives. An engineering budget that buys nodes before it buys a better client-side selector is spending in the wrong order.

(b) Select; do not synthesize. Judge-based selection beats synthesis-style aggregation by 63.1 percentage points, and Mixture-of-Agents-style synthesis loses to the plain single-model baseline in 42 of 42 tasks [39]. This stands in direct tension with MoA's own reported results — 65.1% on AlpacaEval 2.0 against GPT-4 Omni's 57.5% using only open models [41] — and I flag the disagreement rather than choosing the convenient side. The design resolves it

conservatively: selection is the default path, synthesis the exception invoked only at a failed seam.

(c) Heterogeneity is an asset, not a defect. Diverse teams reach 81% win rates where homogeneous teams reach chance, and homogeneous outputs tie 100% of the time — a selector given identical candidates has nothing to select [39]. The protocol should preserve diversity deliberately: dispatch critical fragments to workers of *different* model families, not merely different machines.

The atomicity hypothesis remains unsupported. The strongest form of the claim — that a 3B model answering an atomic sub-task matches a frontier model — is not established. What is supported is narrower, and there is a specific mechanism by which the claim can fail: **a smaller worker requires more context to do the same job**. Knowledge the model does not hold in parameters must be supplied in the prompt. Shrinking the worker is not free; it is paid for in S , and S is paid for in privacy and in cost.

H2 (Capability substitution). For micro-tasks that are atomic, well specified and verifiable, there exists a context budget S at which a 3-8B worker's fragment quality is statistically indistinguishable from a frontier model's on the same micro-task; and the required S decreases as worker capability increases.

H3 (Conversion). A client-side selector over $k > 1$ heterogeneous workers recovers a specified fraction of oracle-selection quality. A recovery fraction that does not improve with orchestrator size would falsify the premise of consequence (a).

An 8B orchestrator is not obviously adequate for that role, and the literature is discouraging: on a stateful planning task, Llama-3.1-8B-Instruct scored near 0-2%, and even frontier models looped in 92-100% of trials when constrained by an external validator [44]. Small models make good *routers* — cheap routers cut costs by over 85% on MT-Bench while retaining 95% of GPT-4 performance [46] — but routing is classification, and planning is not.

7.4 A coverage model for semantic assembly

With the plan as reference sequence, the Lander-Waterman machinery applies — and the source of randomness turns out to be exactly where the model's assumptions are satisfied.

Setup. The plan $\mathcal{D}$ defines an ordered set of semantic units $\mathcal{U} = \{u_1 \dots u_M\}$, fixed before any generation occurs. Each dispatched packet K_i targets a subset $S_i \subseteq \mathcal{U}$. Nominal coverage is $c = (\sum_i |S_i|) / M$.

Where the randomness lives. This is the substantive difference from genomics, and it is what makes the transfer legitimate:

In genome sequencing, the stochastic element is **where the reads land**. In Swarmby, placement is deterministic — the orchestrator chooses it. The stochastic element is **which packets come back**.

Volunteer nodes fail, time out, disconnect and return unusable output, independently and at a rate the network can measure. Let p be that per-packet loss probability. Effective coverage is $c_{\text{eff}} = c \cdot (1-p)$, and the classical results hold with c_{eff} in place of c :

$P(\text{unit } u \text{ is uncovered})$	$= e^{(-c_{\text{eff}})}$
$E[\text{uncovered units}]$	$= M \cdot e^{(-c_{\text{eff}})}$
$E[\text{assembly islands}]$	$= N_p \cdot e^{(-c_{\text{eff}} \cdot \theta)}$

The design equation. Inverting the first result gives a redundancy requirement derived from a stated tolerance:

$$c \geq \ln(1/\epsilon) / (1 - p)$$

Loss rate p	$\epsilon = 5\%$	$\epsilon = 1\%$	$\epsilon = 0.1\%$
0.05	$c \geq 3.2$	$c \geq 4.8$	$c \geq 7.3$
0.10	$c \geq 3.3$	$c \geq 5.1$	$c \geq 7.7$
0.20	$c \geq 3.7$	$c \geq 5.8$	$c \geq 8.6$

And the field numbers to feed it. Anderson & Fedak (2006) [131], over more than 330,000 SETI@home hosts: mean on-fraction **0.81**, connected fraction **0.83**, mean active fraction **0.84**, **host mean lifetime 91 days**. The 2018 version [132] reports ~700,000 devices and availability of ~60% for desktop machines and ~40% for mobile. A loss rate of 16–40% is exactly the regime threshold redundancy is designed for (§12.3).

Scope, and a claim. The model bounds *availability*: the probability that a semantic unit goes unanswered. It does not model semantic correctness — a unit can be covered by five replicas that all agree and are all wrong. Within that scope, I believe this to be **the first coverage model published for semantic assembly**, and it is the point at which the genomic framing stops being a vocabulary and becomes a derivation.

What §6.4 adds to it, and v1.4 did not have. This model answers "will an answer arrive?". It does not answer "does the answer contain the necessary information?". That is Bresler et al.'s information condition, and it is independent: **high coverage with unsolvable fragments produces a confidently wrong assembly**.

7.5 The honest statement of the swarm thesis

The swarm supplies **coverage**; the client supplies **conversion**. Additional nodes raise coverage with logarithmic returns and do nothing for conversion. Heterogeneity among nodes is what makes selection possible, and should be preserved rather than engineered away. The system's quality ceiling is set by the client's selector, and the worker size that suffices is a function of the context budget rather than a constant. And coverage answers only for availability: the solvability of the fragments is a separate condition that no number of replicas repairs.

And now there is a measurement. Over 80 operational table-summary cells (16 tasks × 5 families, equalised output budget), the mean tax per **node class** ranges from **+8.4 %** to **+26.5 %**, with within-class standard deviations of **16** to **39** points (T0RR, §15.9). Under the equalised budget every class comes out expensive — the earlier cheap/expensive split was the output-length confound (T09R) — and the variance *within* a class still exceeds the distance *between* classes. That last part is exactly why per-cell prediction fails (§10.1) and why the swarm answers for coverage rather than for the outcome of any particular cell.

8. Architecture

8.1 Roles

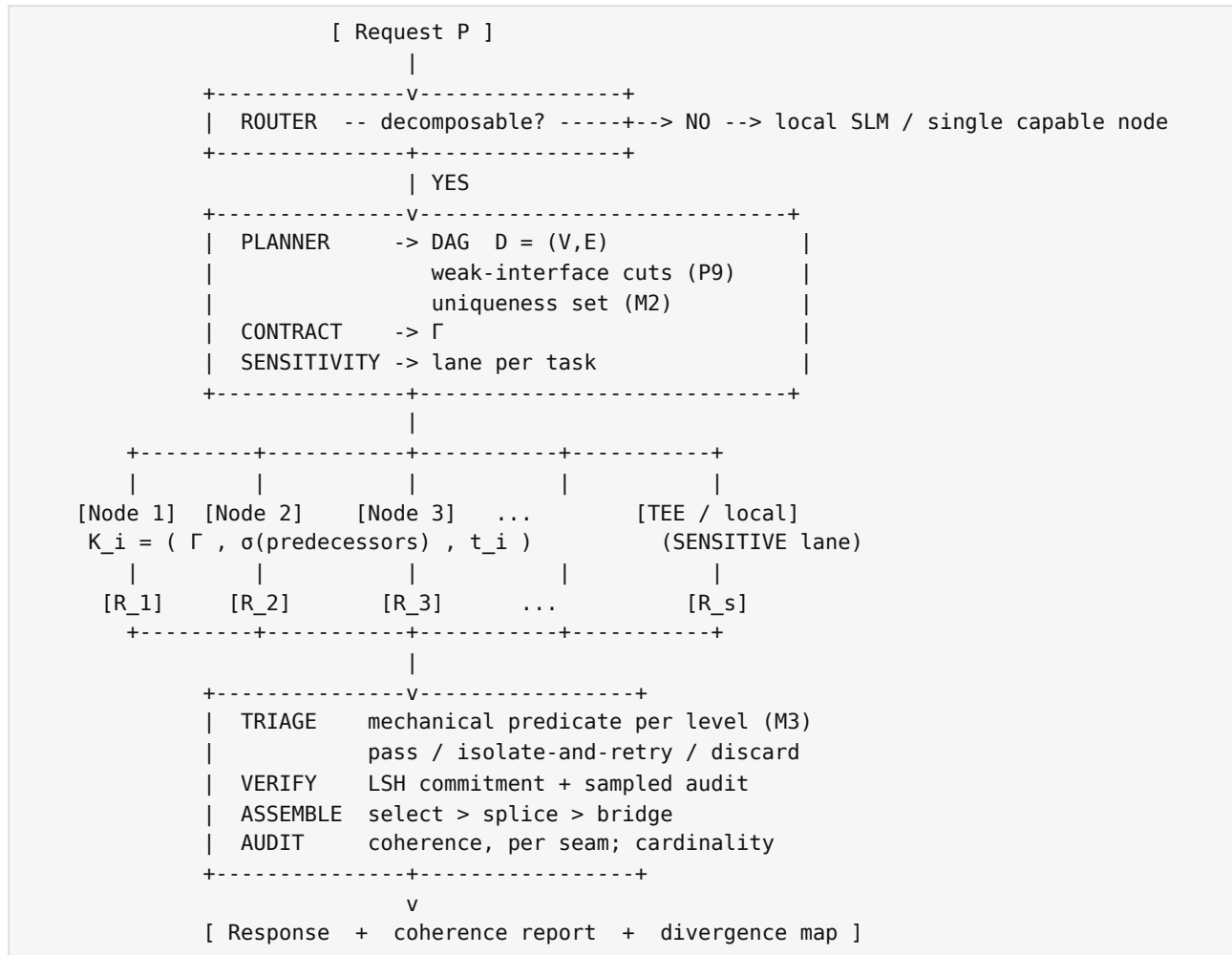
Client / Orchestrator — router, planner, contract generator, sensitivity classifier, packer, speculative dispatcher, verifier, assembler, coherence auditor. Requires an SLM (≥8B

recommended; the adequacy of that figure is H3) plus an embedding model.

Worker node — declares a profile, executes micro-tasks, emits verification commitments and telemetry. Runs one complete small model.

Network services — peer discovery (DHT), reputation registry, credit accounting, audit sampler. Deliberately minimal; no blockchain in v0.3.

8.2 Request lifecycle



Execution proceeds by topological level. Version 1.4 said "a level begins when its predecessors have returned and been verified". This version specifies what that verification means (§10.7) and adds that **consumption may begin before the level is complete**, if the fragment being consumed is complete and has passed triage (§10.7.4).

8.3 Triage, and why it is a specification rather than a metaphor

Biology has a name for that control, and it is precise. Gottesman, Wickner & Maurizi (1997) [116] call it exactly that:

"we propose [...] a general model for what may be thought of as a **triage** system for handling misfolded proteins in vivo, ensuring the **rapid refolding of proteins with functional potential and the rapid degradation of irreversibly denatured or damaged proteins**."

That sentence is almost a specification of the quality-control stage: classify each returned fragment as **recoverable** → **retry** or **unrecoverable** → **discard and regenerate**, and **never**

splice a bad one into the assembly.

The mechanism is equally instructive. The GroEL/GroES chaperonin does not repair the piece in place: it **isolates** it. Xu, Horwich & Sigler (1997) [117] describe how GroES binding "stabilizes a folding chamber" whose elevation and twisting of the apical domains "doubles the volume of the central cavity and buries the hydrophobic peptide-binding residues", leaving a hydrophilic lining "conducive to folding". **Isolate and retry**, not patch in context.

And there is evidence that levels matter, beyond justifying the DAG. Netzer & Hartl (1997) [118] found that two-domain polypeptides "fold efficiently by sequential, co-translational folding" in eukaryotic translation, while the same proteins folded post-translationally in *E. coli* suffer "intramolecular misfolding of concurrently folding domains". Folding the domains one at a time, in order, succeeded where folding them concurrently produced interference. The argument is not that parallelism is bad; it is that **unrestricted** parallelism is. Marsh et al. (2013) [119] add that the assembly order of protein complexes "can be predicted simply from their three-dimensional structures" and that there is **evolutionary selection to preserve order**.

An honest complication. Shiber et al. (2018) [120], via ribosome profiling, found that **nine of twelve** hetero-oligomeric complexes studied assemble co-translationally, and "co-translational assembly often occurs unidirectionally, with a fully synthesized subunit engaging its nascent partner subunit". This complicates the "fold everything, then assemble" picture in an exploitable direction: a level- n fragment can be consumed while level $n+1$ is still being generated, provided the one being consumed is complete. It is an argument for streaming assembly with a completeness condition, not for barrier waiting.

8.4 Latency model

The claim $\max \ll \Sigma$ is directionally right and incomplete. The honest model is

$$T_{\text{total}} = T_{\text{plan}} + \Sigma_{\text{levels}} E[\max_{\{i \in \text{level}\}} t_i] + T_{\text{triage}} + T_{\text{verify}} + T_{\text{assemble}}$$

Two of those terms are local and therefore predictable. Planning scales with $|P|$ and runs on the client's small model; assembly scales with $\Sigma|R_i|$. The other two are set by the swarm.

The first is the straggler tail. $E[\max]$ over W concurrent draws grows with W , and volunteer tails are heavy: the measured effective duty cycle in BOINC is ≈ 0.61 and the median host lifetime is 91 days [47, 48]. With per-node failure probability p , $P(\text{at least one failure}) = 1 - (1-p)^W$; at $p = 0.10$ and $W = 20$ that is 88%. The protocol mandates **hedged requests** at the p95 of the observed per-class latency distribution (§9.6), and this version adds over-dispatch with early termination (§12.3), which attacks the same term by another route.

Triage adds a term v1.4 did not have. It is deliberate and it is cheap: the predicate of §10.7 is mechanical, with no judge and no model. What it buys is that the blast radius of a defective fragment stays contained in that fragment.

8.5 Worker profiles and determinism

A worker that runs an undeclared model, or a different quantization, silently changes fragment quality and register. The profile is therefore part of the protocol and is bound to the verification commitment:

```
profile = (model_family, model_version, quantization,
           prompt_template_id, sampling_params, seed_policy)
```

The orchestrator groups each DAG level into **homogeneous capability classes** to control register mismatch, while deliberately preserving **family diversity across redundant replicas** of the same task (§7.3c). These two goals are in tension and the protocol makes the trade explicit: homogeneity *within* a fragment's role, diversity *across* candidates for the same fragment.

And this version adds $\mathbb{L}$ to the profile. If $\mathbb{L}$ is a property of the node class (§5.7), the node must declare it and the planner must be able to read it, because $\mathbb{N}$ is derived from it. A node that declares an $\mathbb{L}$ it does not sustain produces fragments that are not domains, and that is indistinguishable from the outside from a node that answers badly.

9. Protocol specification (v0.3)

This section is written to be implementable. Field names are normative; encodings are given as JSON for clarity and MAY be CBOR on the wire. Changes relative to v0.2 are marked **[v0.3]**.

9.1 Identifiers

- `session_id` — 128-bit random, generated per request, never reused.
- `task_id` — `BLAKE2b-128(session_id || level_index || task_index)`, truncated to 16 bytes, hex-encoded.
- `attempt_id` — `task_id || ':' || attempt_counter`.

A worker learns `task_id` and `attempt_id`. It **MUST NOT** learn `session_id`; the derivation is one-way so that two workers cannot determine that they hold fragments of the same session by comparing identifiers. This does not defeat timing correlation (§13.2) and is not claimed to.

9.2 Global contract Γ

```
{
  "v": "0.3",
  "objective": "string – what the complete response must accomplish",
  "audience": "string",
  "register": "formal|neutral|informal|technical",
  "format": "prose|markdown|json|code",
  "target_len": 0,
  "lexicon": { "prefer": ["..."], "forbid": ["..."] },
  "entities": [ { "name": "...", "canonical": "...", "role": "..." } ],
  "style_seed": "string – deterministic style anchor shared by all workers",
  "budget": { "max_out_tokens": 0 }
}
```

`entities` is the mechanism that prevents inconsistent naming across fragments — the assembly analogue of a shared coordinate system. `style_seed` is a short fixed phrase all workers are instructed to match, which costs a handful of tokens and materially reduces register drift.

$|\Gamma|$ is the dominant term in S and is therefore the object of the research direction of §6.3. **[v0.3]** It is also the $\mathbb{H}$ term of the derivation in §6.5: measured at ≈ 39 tokens over the project corpus, it is what makes fragmenting too finely expensive independently of coherence.

9.3 Task packet

```
{
  "v": "0.3",
  "attempt_id": "hex",
  "contract": { /* Γ */ },
  "predecessors": [ { "task_id": "hex", "summary": "string", "tokens": 0 } ],
  "task": {
    "instruction": "string",
    "kind": "extract|classify|generate|summarize|transform|judge",
    "expects": { "format": "...", "min_tokens": 0, "max_tokens": 0 }
  },
  "unique_here": [...],
  "flank": { "lead": "string", "trail": "string", "sentences": 0 },
  "constraints": { "temperature": 0.0, "top_p": 1.0, "stop": [...] },
  "commitment_request": { "scheme": "lsh-activation-v1", "params": { "window": 32 } },
  "deadline_ms": 0,
  "lane": "PUBLIC|SANITISABLE",
  "tier": "GLOBAL|TRUSTED",
  "swarm_id": null
}
```

[v0.3] `unique_here` is the list of uniqueness-set elements assigned to *this* fragment and only this one (§11.3). It is not a style instruction: it is the visible half of a global constraint resolved in the plan. A node cannot satisfy "exactly once in the whole output" because it does not know what the others wrote; it can satisfy "these elements go here".

[v0.3] `flank` carries the overlap derived in §6.5, with its length declared in sentences. Declaring it explicitly lets the assembler know which part of the returned material is flank and not body, which is what allows the overlap to be discounted at assembly rather than reappearing as repetition.

Packets on the SENSITIVE lane are never emitted to open nodes. The tier field is orthogonal to lane: it names the *population* a packet may reach rather than the sensitivity of its content.

9.4 Result

```
{
  "v": "0.3",
  "attempt_id": "hex",
  "text": "string",
  "profile": { "model_family": "...", "model_version": "...", "quantization": "...",
    "prompt_template_id": "...", "sampling_params": { }, "seed": 0,
    "L_declared": 0 },
  "commitment": { "scheme": "lsh-activation-v1", "digest": "base64", "bytes": 0 },
  "telemetry": { "gen_ms": 0, "queue_ms": 0, "tokens_out": 0, "energy_j": null },
  "sig": "ed25519 signature over the canonical serialization of all preceding fields"
}
```

[v0.3] `L_declared` is the fragment size the node class sustains (§8.5). The orchestrator uses it to derive N and to detect nodes whose declared L does not match their observed behaviour.

9.5 Node profile advertisement

```
{
  "node_id": "ed25519 public key",
  "models": [ { "family": "...", "version": "...", "quantization": "...",
    "ctx": 0, "tok_per_s_est": 0, "L_sustained": 0 } ],
  "capabilities": { "tee": false, "attestation": null },
  "swarm": { "swarm_id": null, "registry": null, "mtls_cert_fingerprint": null },
  "resources": { "vram_mb": 0, "ram_mb": 0 },
  "policy": { "max_tokens_per_task": 0, "kinds": ["extract","generate"] },
  "reputation": { "completed": 0, "audit_pass_rate": 0.0, "since": "ISO-8601" }
}
```

9.6 Dispatch, hedging, over-dispatch and retry

1. Filter candidates by tier first — a packet marked TRUSTED is offered only to nodes whose public key appears in the named swarm's whitelist — and then by declared capability, kind support, and observed RTT.
2. For a task of criticality k , dispatch to k nodes selected to **maximize model-family diversity** subject to the capability class.
3. Start a hedge timer at the **p95** of the observed latency distribution *for that task kind and token budget*, not at a fixed constant. On expiry, dispatch an additional replica; accept the first result that verifies.
4. **[v0.3] Over-dispatch with early termination.** For a level of N distinct sub-tasks, dispatch $N + m$ and accept the first N that return and pass triage, discarding stragglers. **Normative condition:** this is admissible only when the m extras are *genuinely interchangeable* sub-tasks (§12.3). If the content of fragment j is needed, receiving the other $N-1$ does not substitute for it, and the scheme does not apply.
5. Cancel outstanding replicas on acceptance. Record cancellations — a node whose work is habitually cancelled is slow, not dishonest, and reputation must distinguish the two.
6. On verification failure, re-dispatch excluding the failing node and record the event for the audit sampler.

9.7 Parameter table

Parameter	Symbol	Default	Derivation
Fragment size	L	property of the node class	§5.7; declared by the node, verified by behaviour
Fragment count	N	derived	$N = \lceil \text{material}/L \rceil$ (§5.7). No longer a parameter
Flank	F	measured	$\max(\text{longest repeat, longest crossing dependency})$ (§6.5)
Header cost	H	measured	≈ 39 tokens on the project corpus (§6.5)
Context budget	S	derived	$\backslash S =$
Redundancy ratio	ρ	derived	$(L+2F)/L + H/(L \cdot s)$ (§6.5). Reported, not configured
Dependency density	δ	measured	necessary relations crossing a boundary, per fragment (§6.8)
Seam threshold	τ_{sem}	calibrated	§10.6; never a constant
Router threshold	τ_{route}	calibrated, asymmetric	$\beta < 1$ in F_{β} [12]
Criticality replicas	k	split into three	§12.4: $k_{\text{avail}}, k_{\text{verif}}, k_{\text{epist}}$
Hedge trigger	—	p95 per class	§8.4
Over-dispatch extras	m	0 unless interchangeable	§9.6.4, §12.3

Parameter	Symbol	Default	Derivation
Audit sampling rate	λ	0.01–0.05	§13.3
Max plan width	—	8	The straggler tail grows with width
Max plan depth	—	4	Beyond this, dependency density argues for not fragmenting

10. Algorithms

10.1 Router

```
function is_decomposable(P) -> (bool, score)
  f ← features(P):
    · task-kind signals (extract / enumerate / summarize vs prove / derive / refactor)
    · length of P
    · density of sequential-dependency markers
    · presence of shared mutable state (code, ledgers, running totals)
    · request for a single artifact vs a set of items
    · [v0.3] estimated dependency density  $\delta$  (§6.8)
  score ← classifier(f)
  return (score >  $\tau_{\text{route}}$ , score)
```

τ_{route} is calibrated with **F_β**, $\beta < 1$: a false positive — fragmenting something that should not be — costs more than a false negative. That is the SoT-R lesson, and it is the difference between a system that improves on 2023 and one that regresses from it.

[v0.3] Including δ as a feature is a direct consequence of §6.8: if distortion depends on dependency density and not only on rate, then the router — which is what decides whether to fragment — must see that density. A router that sees only length is measuring the wrong axis.

And that feature was measured, with a result that forces a design change. Over 80 operational cells, a classifier trained with leave-one-task-out validation reaches **AUC 0.57** with δ and reputation, and **0.61** with neither: adding δ improves nothing (T0RR). With pre-dispatch capability probes in place of δ , **AUC 0.57** (T0RR2). **Per-cell prediction does not work**, and the failure is not δ 's in particular: no subset of features rescues it.

What does separate is the **node class**. The same corpus gives mean taxes per family from +8.4 % to +26.5 %, and a policy that decides by class — fragment if that class's historical tax is negative — obtains **+0.0 %** against **+14.8 %** for always fragmenting: under the equalised budget every class has a positive historical tax, so the class policy refuses them all and avoids the whole tax. The per-class policy is the router that works: it needs no per-cell prediction, and its decision is exactly the one P2 prescribes when the measured economics say "do not fragment".

Design consequence. The router of §10.1 decides *whether the prompt is decomposable* — and that remains verified: 22 of 22 decisions agree with the corpus truth (T0R). What it must **not** attempt is to predict the cost of a particular cell. That decision moves to the node profile: the class declares its history, and the orchestrator routes by class. δ is kept as a feature of the plan and **withdrawn as a predictive feature of the router**.

10.2 Planner

```
function plan(P, L) -> (D = (V,E), Uniq)
  boundaries ← candidate_cuts(P)                # [v0.3] P9
  weight     ← estimated_coupling(boundary)      # dependencies that would cross
  cuts       ← select_minimal(boundaries, weight, size_target = L)
  units      ← segment(P, cuts)
  for each ordered pair (u, v):
    E ← E ∪ {(u,v)} if v requires the *result* of u, not merely its statement
  assert acyclic(D)
  if width(D) == 1: return REFUSE                # a chain is not parallelizable
  if depth(D) > max_depth: return REFUSE
  Uniq ← uniqueness_set(P, Γ)                   # [v0.3] M2, §11.3
  assign each element of Uniq to exactly one node of V
  return (D, Uniq)
```

The distinction between requiring a predecessor's *result* and requiring its *statement* is the crux. Least-to-most prompting achieves $\geq 99\%$ on SCAN against 16% for chain-of-thought precisely by respecting result-dependencies sequentially [51]; a planner that mistakes a result-dependency for a statement-dependency converts that gain into a loss.

[v0.3] Two substantive changes. First, the cut is no longer by size: L enters as `size_target` and the selection minimises the coupling cut, which is P9 and is the minimisation of δ from §6.8. Second, the planner computes the uniqueness set and assigns it — the global constraint is resolved here, where global information exists, and not at the node, which has none.

10.3 Packing

```
function build_packet(Γ, t_i, preds, L, F) -> K_i
  base ← Γ                                     # never trimmed
  flank ← take_flank(neighbours(t_i), F)       # [v0.3] derived, not a percentage
  order preds by (edge weight, recency)
  for p in preds while budget remains:
    s ← summarize(p.result, min(budget, cap_per_pred))
    attach(s)
  return (Γ, flank, attached, t_i, unique_here(t_i))
```

Γ is never trimmed to meet a budget. If the budget falls short, the planner **increases L** — fewer, larger fragments — rather than trimming shared context. Version 1.4 said "reduce N ", which is the same thing under the old formulation; under the new one, N is derived and L is the lever. It is the LongRAG result, where 4K-token retrieval units and fewer than eight top units matched fully trained state of the art with no training at all [52].

10.4 Assembly

```
function assemble(fragments, D,  $\Gamma$ , Uniq,  $\tau_{\text{sem}}$ ) -> (text, seam_report, cardinality_report)
  ordered  $\leftarrow$  topological_flatten(D)
  chosen  $\leftarrow$  []
  for t in ordered:
    cands  $\leftarrow$  fragments[t]
    chosen.append( cands[0] if |cands| == 1
                  else judge_select(cands,  $\Gamma$ ) )      # select, do not synthesize
  discount_flanks(chosen)                             # [v0.3] §9.3
  out, seams  $\leftarrow$  [], []
  for (a, b) in consecutive(chosen):
    sim  $\leftarrow$  cos( embed(tail_window(a)), embed(head_window(b)) )
    if sim  $\geq$   $\tau_{\text{sem}}$ :
      out.append(a); seams.append((a,b,"splice",sim))
    else:
      bridge  $\leftarrow$  slm.write_transition(tail(a), head(b),  $\Gamma$ )
      out.append(a); out.append(bridge)
      seams.append((a,b,"bridge",sim))
  card  $\leftarrow$  verify_cardinality(out, Uniq)              # [v0.3] M2, §11.3
  return join(out), seams, card
```

Every seam is recorded with its similarity and the path taken. The seam report is part of the response, per P6.

[v0.3] Two steps are added. `discount_flanks` removes the declared overlap before splicing, so that the flank does its job — enabling repeat detection and continuity — without reappearing as duplicated text. `verify_cardinality` checks the output against the set computed in the plan, not against a rule stated in the contract.

10.5 Consensus by multiple alignment of replicas (E16, revised)

Section 10.4 resolves *different* fragments in *different* positions. This section resolves *k replicas of the same micro-task*.

Two levels, deliberately distinct.

Level	Unit	Mechanism	When
Macro	Different sub-tasks of one large task	Overlap-and-splice with flank (§10.4)	Long generative work
Micro	<i>k</i> complete replicas of the same micro-task	Multiple alignment and consensus (this section)	Every micro-task of criticality <i>k</i> > 1, including a request that was atomic from the start

An atomic request — one that the router declines to decompose — skips the macro level entirely and goes straight to the micro level with *k* replicas. **Splitting an atomic question into partial questions is not a supported operation**, because it removes information before sampling rather than sampling redundantly; no amount of coverage recovers it.

10.5.1 What the state of the art does, and where the opening is

The recent literature on uncertainty in generation is abundant and good.

Semantic entropy (Kuhn, Gal & Farquhar, ICLR 2023 [98]; Farquhar et al., *Nature* 2024 [99]) samples several answers, clusters them by semantic equivalence via **bidirectional entailment**, and computes entropy over meaning clusters instead of over token sequences. Average AUROC 0.790 across 30 model-and-task combinations.

SelfCheckGPT (Manakul, Liusie & Gales, EMNLP 2023 [100]) produces **per-sentence** factuality scores by comparing each sentence against complete samples.

Semantic agreement across different models (Soiffer et al., 2025 [101]) uses agreement among smaller models as a deferral signal.

What all of them share, and this is the opening: they treat the k answers as a **bag to be clustered**. None treats them as **sequences to be aligned**. Clustering yields a scalar per answer; alignment yields a **position-to-position** correspondence, which is what allows one to say *where* they diverged and not only *how much*.

10.5.2 The instrument exists and is mature

Substitution cost is learned, not postulated. That is the lesson of BLOSUM (§5.2). In the text domain it is already solved at the string level: Ristad & Yianilos (1998) [96] give "an efficient algorithm for learning the primitive edit costs from a corpus of examples", via a stochastic transducer fitted by EM.

And it is solved at the phrase level. PPDB 2.0 (Pavlick et al., ACL 2015) [97] is a paraphrase database rescored discriminatively: they collected human judgements over **26,455 pairs**, each rated by 5 people on a 5-point Likert scale, and fitted a ridge regression over **209 features**.

The result is the single most useful datum in this whole investigation for the project:

The heuristic ranking correlated with human judgement at $\rho = 0.41$. The empirically fitted model reached $\rho = 0.71$. And in that model embedding cosine is one feature out of 209.

That is published, quantified evidence that **raw embedding distance is not the right instrument** for judging semantic interchangeability — which is exactly what the map needs to judge, and exactly what v1.4 used.

Consistency-based alignment is the design this wants. T-Coffee (Notredame, Higgins & Heringa, 2000) [94] builds a primary library of pairwise alignments and then **extends** it: for each pair of residues it examines their alignment with residues from the other sequences, and "the weight associated with a pair of residues will be the sum of all the weights gathered through the examination of all the triplets involving that pair". Translated: **an agreement corroborated through a third, independent answer weighs more than an agreement between two**. And the progressive phase uses "position-specific scoring" instead of a fixed matrix.

And the formalisation of the product exists too. A profile HMM (Eddy, 1998) [95] "turns a multiple sequence alignment into a position-specific scoring system". A position-specific scoring model built from a set of aligned answers is the formal description of a confidence map.

10.5.3 The proposed model

M1 — Multiple alignment of answers with learned semantic substitution.

1. **Segment** each of the k answers into sentences (the alignment unit of §5.7).
2. **Align** the k sentence sequences by dynamic programming with affine gap penalties (Gotoh, 1982) [93], using a semantic substitution matrix.
3. **Build that matrix empirically**, BLOSUM-style: count pairs of sentences that appear as mutual substitutions in cases verified as correct, and score $\log(q_{\text{obs}} / e_{\text{exp}})$. The

embedding enters as a feature, not as a verdict.

4. **Extend by consistency**, T-Coffee-style: weight each correspondence by its corroboration through the other answers.
5. **Emit** a position-specific profile over the resulting alignment. That is the map.

What is new and what is not. Nothing in steps 2, 3 and 4 is invention: they are Needleman-Wunsch with Gotoh, Henikoff with Ristad, and Notredame. What is new is **applying them to outputs from different models instead of to biological sequences**, and doing so to produce a per-unit annotation rather than a scalar per answer. The contribution is the assembly of the instrument, not its parts, and it must be stated that way.

10.5.4 What remains withdrawn

The confidence map's reliability claim remains withdrawn, and M1 does not lift it.

Version 1.4 measured the correlation between per-unit agreement and correctness, first against a peer-class judge ($r = -0.030$ over 597 units, with the judge accepting 93.3% — too saturated to discriminate) and then against an answer key in three runs, which returned common odds ratios of **3.47, 0.26 and 1.24**: above, below and astride 1 on the same question. Three mutually contradictory estimates are not a weak signal; they are no signal, measured three times.

M1 proposes a **different instrument**, not a reinterpretation of that result. That means three things, and they belong together:

1. Confidence labels are reported as **agreement**, never as **accuracy**, and the map is not offered as a reliability guarantee. This does not change.
2. **The mechanism is still real.** Reporting *where* independent replicas diverged is a capability, and it is the one a single-model provider does not have. What may not be presented is convergence as evidence of correctness.
3. M1 has its own death condition, stated in §15.6: if with the correct instrument and **in a regime where the models genuinely disagree** the per-unit agreement does not beat chance, then localisation creates no signal and M1 falls. The regime clause is not an escape hatch: it is the correction of the known defect in the first measurement, where the judge accepted almost everything and no variance remained for a correlation to appear against. Declaring it in advance is what prevents it being used afterwards.

Three honest caveats that do not change. Agreement is not truth: models trained on overlapping corpora share errors, and convergence on a common falsehood is a correlated failure that alignment cannot see — which is why §9.6 mandates cross-family diversity. The map costs $k\times$, and it is applied by criticality, not universally.

10.6 Threshold calibration

```
function calibrate_tau(labelled_pairs, embedder,  $\beta = 0.5$ ) -> ( $\tau^*$ , curve)
  sims  $\leftarrow$  [ cos(embed(a.tail), embed(b.head)) for (a,b,label) in pairs ]
  for  $\tau$  in quantiles(sims, 200):
    compute precision/recall of "is a broken seam"
     $F_\beta \leftarrow (1+\beta^2) \cdot P \cdot R / (\beta^2 \cdot P + R)$ 
  return argmax_ $\tau$   $F_\beta$ , curve
```

$\beta < 1$ weights precision: declaring a seam broken triggers a rewrite, and unnecessary rewrites are how a system degrades text that was already fine. τ must be re-derived whenever the embedding model changes, for the anisotropy reasons of §6.9.

[v0.3] While M1 is not built, this threshold is the operative instrument and is retained. When it is built, the learned substitution matrix replaces it at the micro level (§10.5) and τ_{sem} is restricted to the macro level, which is where it compares *the end of one fragment with the start of the next* rather than two versions of the same thing.

10.7 The triage gate

```
function triage(carry, task_that_requested_it) -> PASS | RETRY | DISCARD
  if not answers(carry, task_that_requested_it):    # mechanical predicate
    return RETRY if attempts < r else DISCARD
  if violates_schema(carry, task.expects):
    return RETRY if attempts < r else DISCARD
  return PASS
```

M3 — Triage gate per topological level.

1. A level does not start until the carries from its predecessors have passed a **mechanical predicate**: *does the carry that arrived answer the task that requested it?* No judge, no model, cheap.
2. A fragment that does not pass is **isolated and retried**, not patched in context nor spliced in with a warning.
3. A fragment that does not pass after r retries is **discarded and regenerated** with a different plan. Degradation is a legitimate output of triage, not a system failure.
4. Consumption may begin **before the level is complete**, if the fragment being consumed is complete and verified (§8.3, Shiber et al.).

Why it matters, with a datum from the project. There is a recorded incident this gate catches: 42 of 60 dispatched packets carried another packet's answers. Without a gate, carries pass forward unreviewed and one level's error multiplies into the next. With a gate, the blast radius is one fragment.

And its death condition. If the gate rejects so much legitimate work that the retry cost exceeds the damage it prevents, M3 falls. It is a cheap measurement and its prediction is binary: the incident happens or it does not.

10.8 Coherence audit

Two instruments, reported separately and never merged into a single "quality" score — merging is exactly how the damage hides [12]:

1. **Entity-grid local coherence** [53] — entity mentions and grammatical roles across sentences, scored from transition probabilities.
2. **Seam error taxonomy** — the mechanically detectable subset of the error classes identified from 1,193 human annotations over 100 books [54]: entity omission, duplicated content, contradiction, register or tense shift, dangling reference, missing transition, repeated introduction, inconsistent naming.

The headline figure is the **coherence tax**: relative degradation against monolithic generation with the same model.

[v0.3] A third report is added, separate from the other two: the **cardinality report** (§11.3, point 4), with over-compression and over-expansion counted separately. They are different errors with different causes, and measuring them together hides both.

And a limitation v1.4 documented that still stands. The entity grid does not function on short answers: it returned monolithic baselines between 0.000 and 0.114 across the whole corpus, which makes every relative comparison built on it a ratio over a near-zero denominator. Either the evaluation corpus moves to longer outputs or the instrument is replaced; until then this document has one working coherence instrument, not two (L14).

11. Cardinality constraints

This section is new. It treats the class of constraint no node can satisfy in isolation: "mention this term exactly once", "do not repeat a formulation". It is, by the project's measurements, the irreducible class.

11.1 A hypothesis that did not survive

The hypothesis was: *mate pairs are the homologue of global constraints, because they are the mechanism genomics invented for information that no individual read contains.*

Background concept. A *mate pair* or paired-end read is a pair of reads sequenced from the two ends of the same DNA fragment of approximately known length. Weber & Myers (1997) [106] introduced it because "read pairs from both ends have known spacing and orientation", which "aids the assembly of sequences containing dispersed repetitive elements".

Half the hypothesis is correct: constraints that cross fragments are real, they are the central difficulty, and genomics did invent explicit machinery for information that no single piece contains.

The other half is wrong, and it is wrong along four axes at once:

axis	mate pair	"exactly once in the whole output"
Arity	Binary and pre-identified: it names two concrete reads, and the pair exists before assembly because library preparation created it	n-ary and quantified: it names no pair; it quantifies over all fragments
Metric content	It is fundamentally a distance , with a distribution — Myers et al. (2000) [107] report insert lengths "normally distributed with 10% variance"	It has neither distance nor orientation
Hardness	Soft: Opera (Gao, Sung & Nagarajan, 2011) [108] defines concordance as a predicate the optimiser maximises ; a discordant mate pair is tolerated	Hard: a single violation is a failure, not a datum to be outvoted
Direction of information	Evidence: an additional observation sampled from a genome that already exists	Specification: there is no true answer to recover; it is a condition imposed on the output

The fourth axis is the deepest. Genome assembly is **maximum-likelihood inference toward a correct answer that pre-exists**. Swarmably does **constraint satisfaction over a space of acceptable outputs**. Mate pairs live on the inference side.

What a mate pair actually is. The transfer is not null: it is narrower in scope. Mate pairs are the correct homologue of a real class of constraint: **"fragment A and fragment B must be mutually consistent at a known relative position"**. The conclusion must pick up the anecdote from the introduction; section 4 must resume the thread left open by section 2; a reference to a figure in one fragment must correspond to a figure defined in another. They are binary, they have a positional component, they are known in advance because the plan

created them, and they degrade gracefully when partially violated. For those, the argument of §11.3 transfers along with the homology.

11.2 The correct homologue is in the same literature

Genomics does have a mechanism whose form is "this must appear exactly N times in the whole output". It is not mate pairs. It is **multiplicity** and **uniqueness**.

k-mer multiplicity. Compeau, Pevzner & Tesler (2011) [104] state it as a counting operation that enters the graph structure: one must determine "how many times each k-mer appears", and "if the multiplicity of a k-mer is m, we will connect its prefix to its suffix using m directed edges (instead of just one)". That is a global cardinality constraint imposed **structurally** rather than as an after-the-fact check.

U-unitigs. Myers et al. (2000) [107], in the *Drosophila* assembly, describe that those units "that are certain to represent unique DNA were designated U-unitigs" — and all the scaffolding is anchored on them. It is literally the "exactly once in the whole output" mechanism: the assembler **computes the set of sequences that must appear exactly once** and builds around them.

The failure mode, which confirms the transfer. Myers (1995) [105] names the pathology when this is done badly: seeking the shortest string containing all fragments means that "in the case of repetitive target sequences this objective produces over-compressed answers". **Over-compression** is exactly the failure of an assembler that merges two passages that should have remained distinct. And its dual — emitting twice something that should have appeared once — is the failure `no_repeated_ngram` measures.

11.3 The proposed model

The strongest architectural lesson comes from Medvedev et al. (2011) [109], who argue against after-the-fact treatment: mate pairs have been incorporated "as various heuristic post-processing steps", which "may still fail to resolve complex repeats"; their proposal is to incorporate the information **into the graph structure itself**.

M2 — Cardinality constraints resolved in the representation, not in the output.

1. **Compute the uniqueness set before dispatch.** Which terms, entities and formulations must appear exactly once in the final output. It is the analogue of identifying U-unitigs.
2. **Assign each element of that set to exactly one fragment**, as a property of the plan and not as an instruction to the nodes. A node cannot satisfy "exactly once" because it does not know what the others wrote; the planner can. It is the `unique_here` field of §9.3.
3. **Verify cardinality at assembly**, against the set computed in step 1, not against a rule stated in the contract.
4. **Log the two failure modes separately:** over-compression (what should have been distinguished was merged) and over-expansion (what should have appeared once was repeated).

Point 2 is the real change, and there is a number to compare it against. Today the project asks for `term_once` in the contract and enforces it mechanically at the assembler. That works, and the measurement confirms it: mechanical enforcement raised compliance from **6/24 to 18/24**, above the monolithic arm's 13/24. But it is an after-the-fact correction.

Assigning uniqueness in the plan makes the constraint impossible to violate rather than detectable afterwards, which is precisely Medvedev et al.'s argument.

Its death condition, stated in advance: if assignment in the plan does not reduce violations below what mechanical enforcement at the assembler already achieves (18/24), M2 adds nothing and falls. M2's prediction is strong — the violation rate should fall to zero, not improve — precisely so that it is easy to refute.

It was measured, and the strong prediction is false as written. Over 35 unique terms in 10 real cells (2 tasks × 5 families), nodes that receive `unique_here` produce **6 of 35** exactly once: **0 terms omitted** and **29 repeated** (T03R). A small model does not obey "exactly once" because it is told to; the field tells it what is its share, not how many times to write it.

And yet M2 stands, corrected. Running those same outputs through the assembler's mechanical enforcement — the v1.4 one, not a new one — the result is **35 of 35** exactly once. The asymmetry is the finding: over-expansion is trimmed mechanically, over-compression is not invented, and there were **zero** omissions. That is: **assignment in the plan removes the irreparable failure mode, and enforcement at the assembler removes the reparable one.**

Correction to M2. The mechanism is not "assignment in the plan **instead of** enforcement at the output", but **assignment in the plan plus mechanical enforcement at assembly**. Assignment *localizes* responsibility; enforcement *guarantees* it. The phrase "zero by construction" is **withdrawn**: what is zero by construction is over-compression, not cardinality violation in general. The declared death condition (beat 18/24) is met — 35/35 over 35 terms — but it is met thanks to a component M2 proposed to render unnecessary, and that has to be said in those words.

12. Redundancy: what k is for

12.1 A second hypothesis that did not survive

The hypothesis was: *instead of sending the same packet to k nodes, send $N + m$ encoded fragments of which any N suffice to reconstruct.*

The motivation was good and is well founded in the storage literature. Weatherspoon & Kubiatowicz (2002) [126] show that erasure codes "use an order of magnitude less bandwidth and storage than replication for systems with similar MTTF": with a million machines and 10% down, two full replicas give "only two nines of availability", while an encoding into 32 fragments gives "more than eight nines" with the same storage. Fountain codes take that to the limit: Luby (2002) [124] proves recovery from "any $k + O(\sqrt{k} \cdot \ln^2(k/6))$ of the encoding symbols with probability $1 - \delta$ "; Shokrollahi (2006) [125] improves it to "any subset of symbols of size $(1+\epsilon)k$ " with $O(1)$ operations per symbol.

Why it does not transfer. The reason is clean and admits no nuance.

Fountain codes are linear codes over shared unknowns. Shokrollahi's definition says so: "each output symbol is the sum of some of the input symbols", and decoding is Gaussian elimination or peeling over that linear system. The entire mechanism by which "any $k(1+\epsilon)$ suffice" **is** that the received symbols are **linear equations in the same unknowns**.

Swarmbly's fragments are not equations in shared unknowns. Each node **generates new text**. There is no XOR, no algebraic field, no inverse. There is nothing to solve for.

And this is not an inference of our own: it is the explicit limitation stated in the coded-computing literature. Kosaian, Rashmi & Venkataraman [128] note that much work employs erasure codes for computing **linear functions**, and that "to the best of our knowledge, none of the existing works are applicable to broader classes of non-linear computations". What does exist confirms the diagnosis from the positive side: Mallick et al. (2019) [127] obtain up to **3× speedup** applying rateless codes to distributed matrix-vector multiplication — **because matrix-vector multiplication is linear**. And the only serious attempt at coding non-linear inference, ApproxIFER [129], achieves **approximate** recovery over low-dimensional classification outputs, with accuracy losses of up to ~6-9% in degraded mode.

A specific search was made for prior work applying fountain codes to LLM agent redundancy or to decentralised LLM inference. **None exists**. The transfer has not been attempted, and linearity explains why.

12.2 Applying the method of §3

This fall is the cleanest example of the failure-mode test. Fountain codes were proposed for their recovery property — "any $k(1+\epsilon)$ suffice" — with no condition on when that property ceases to exist. The condition exists, it is published, and it is linearity. Looking for it would have saved the entire hypothesis.

12.3 What does transfer: the economics, not the mechanism

The real lesson of the storage literature is not about XOR. It is about **thresholds and quorums**: for a target reliability, $(N, N+m)$ beats k -fold duplication.

M5a — Over-dispatch with early termination. Dispatch N distinct sub-tasks plus m extras, accept the first N that return and pass triage, discard stragglers. It is a threshold scheme, it delivers the ρ saving that was being sought, **and it is not a fountain code**. Calling it one would be the very error Section 3 exists to avoid.

With one condition that must be stated, because it is the core of the trade-off: the m extras are only free if the sub-tasks are **genuinely interchangeable**. The entire point of erasure coding is that *any* k symbols serve. Swarmby's fragments are not interchangeable: if the content of fragment j is needed, receiving the other $N-1$ does not substitute for it. Making them interchangeable requires content redundancy, which costs exactly the ρ one wanted to save. That is why §9.6.4 makes it normatively conditional.

12.4 Separating the three purposes of k

This is the finding that reorders the role of k , and it is probably more valuable than the original hypothesis.

k today serves **three distinct purposes** that the design does not separate:

- **Availability** (k_{avail}) — that the task completes even if a node goes down.
- **Verification** (k_{verif}) — that a dishonest node cannot impose a false result.
- **Epistemic redundancy** (k_{epist}) — that the divergence map has something to align.

Sarmenta (2002) [133] measures the cost of each route in volunteer computing: voting "reduces error rates exponentially with redundancy, but requires all work to be done several times, and does not work well when there are many saboteurs"; spot-checking "reduces the error rate linearly with the amount of work to be done, while only costing an extra fraction of the original time"; and the credibility-based combination yields "mathematically guaranteeable levels of correctness with much less slowdown". BOINC [130] implements the

deployed version: adaptive replication achieving "a low bound on the error rate [...] even in the presence of malicious volunteers, while imposing only a small performance overhead".

M5b — The design consequence. If k exists for **verification**, erasure codes were never the tool; credibility-based spot-checking is. If k exists for **availability**, threshold over-dispatch solves it. And if k exists for the **divergence map** — epistemic redundancy — then it is replaceable by neither, because its product is not fault tolerance but signal.

Three purposes, three mechanisms, and today a single parameter. **Separating them is an immediate improvement to the protocol, available without measuring anything** — which is why §15.6 puts it first in the order of attack.

And it connects with something v1.4 had already discovered by another route. The trusted-swarm section (§13.5) observed that a cryptographic whitelist removes the need for adversarial redundancy but not for epistemic redundancy, and required that lowering k to 1 be declared explicitly. That was this separation, seen from a particular case. Now it is general.

Its death condition: if the three turn out to be so coupled in practice that separating them saves nothing, M5 falls.

13. Privacy, verification and adversarial nodes

13.1 Fragmentation is not encryption

Early development of this concept described decontextualized fragmentation as a form of "quasi-encryption," on the reasoning that a node holding one fragment without global context holds nothing of value. That reasoning does not survive contact with the re-identification literature, and the claim is withdrawn.

Four findings converge. The first is that quasi-identifiers suffice: the combination of ZIP code, date of birth and gender uniquely identifies roughly 87% of the US population even though none of the three is an identifier on its own [55], and although a later revision puts the figure near 63%, that is not reassuring. Every syntactic fragmentation or generalization defence proposed in that literature has since been broken by a subsequent attack [56], and sparse high-dimensional data turns out to be inherently re-identifiable from a handful of coarse, noisy attributes [57].

The second is that style is itself an identifier. Authorship attribution operates at internet scale [58], survives shortening and cross-platform domain shift [59], and functions below 280 characters [60]. The third is that intermediate representations invert: text embeddings reveal almost as much as the text itself [61], and prompts can be recovered from model outputs alone [62].

The fourth is decisive because it addresses this architecture directly. In split inference, the ActInv attack achieves precision and recall above 98% in nearly all evaluated cases, with ROUGE-L consistently above 0.96. Cutting after two client blocks of Qwen3-0.6B yields 99.76% precision, and even at seven blocks it retains 77.74%. Defences underperform: at 70% activation sparsification precision decreases "only modestly" [63].

Swarmbly does not transmit activations, which places it in a better position than split inference. But the direction of the evidence is unambiguous, and there is a further argument internal to this design: **§6.2 establishes that coherence requires shipping the global contract Γ to every worker.** A node holding Γ holds the objective, audience, format and

constraints of the session. Decontextualization and coherence are antagonists by construction.

13.2 What can be claimed

Swarmbly reduces the exposure surface relative to a centralized provider that reads and retains the complete prompt, and relative to pipeline-parallel schemes in which nodes observe intermediate activations and text under generation. It provides **no cryptographic guarantee of confidentiality**. An adversary controlling a significant fraction of nodes, or correlating by timing and session identifier, can reconstruct a substantial portion of a session.

Two residual channels are worth naming because they are cheap to overlook: **timing correlation** (fragments of one session arrive in a burst) and **contract fingerprinting** (a distinctive Γ is itself a session identifier across the nodes that receive it). Mitigations — jittered dispatch, per-node contract paraphrase — cost latency and coherence respectively, which is again §6.2.

[v0.3] And this version adds a third channel, as a consequence of M2. The `unique_here` field (§9.3) tells a node which elements are unique across the whole output. That is global information about the session which v1.4 did not send. The trade is explicit: M2 buys cardinality correctness at the price of leaking the document's uniqueness structure. On the SANITISABLE lane this is acceptable; anything above it, `unique_here` must be omitted and cardinality verification must fall entirely to the assembler — which is v1.4's behaviour and is worse, but is the right behaviour there.

13.3 Verification

Strong cryptographic confidentiality is unaffordable here, and it is worth stating the numbers so the conclusion is not mistaken for defeatism. General-purpose MPC on a transformer runs at a slowdown of order 10^4 – $10^6\times$, with 280.99 GB of communication for a single BERT-Base inference [64, 65]; the best two-party systems report roughly 8 minutes per token for LLaMA-7B [66]. Zero-knowledge proofs of inference need under 15 minutes to prove one forward pass of a 13B model [67]. None of this fits a volunteer economy.

What does fit is a layered scheme:

Layer 1 — computational integrity by locality-sensitive commitment. A commitment scheme over activations detects unauthorized model, prompt or precision substitution with 100% accuracy, zero false positives and zero false negatives in reported testing, at 258 bytes per 32 tokens — roughly $1000\times$ compression against raw embeddings — validating faster than the original inference [68]. This is what makes a node market possible: it costs almost nothing and it closes the obvious fraud, which is a node advertising an 8B model and serving a 1B one.

Layer 2 — sampled public audit. Verification at approximately 1% of inference cost, secure under a *one-honest-verifier* assumption rather than an honest majority, with failure probability $P_{\text{fail}} \sim \rho^k$ [69]. The essential design property: **workers cannot distinguish an audit task from a real one.**

Layer 3 — selection as a defence. With $k > 1$ replicas, the judge-based selection of §10.4 already discards anomalous fragments as a side effect of improving quality [39]. This is the cheapest defence in the system because it is paid for by something else.

[v0.3] Layer 0 — triage. The gate of §10.7 is not a cryptographic defence and is not presented as one, but it acts first and it is free: a fragment that does not answer the task that requested it does not enter the assembly, whether it comes from a slow node, a broken one or a dishonest one. Its value is containment, not detection.

What none of these layers does is verify *semantic faithfulness*. Layer 1 proves that a declared model was run on a declared input; it does not prove that the resulting prose is true, or non-malicious. That gap matters because every returned fragment is untrusted input flowing into the client's model — squarely the territory of prompt injection, supply chain and improper output handling [70]. And the client cannot be relied on to notice: off-the-shelf reasoning models attribute failures in agentic systems at under 10% accuracy [71]. A system that cannot attribute *honest* failures will not detect adversarial ones. The protocol's answer is to constrain the blast radius — fragments are data, never instructions; the assembler runs with output-handling defences; and kind-specific output schemas are enforced before a fragment enters the assembly context.

13.4 Sensitivity lanes

Lane	Criterion	Destination	Cost
PUBLIC	No PII, no commercial secret	Open volunteer nodes	None
SANITISABLE	Detectable, pseudonymizable PII	Open nodes; rehydrated locally	Real residual risk (below)
SENSITIVE	Health, legal, financial, identifiable	Local execution, or attested TEE	<7% mean overhead on H100 confidential computing [72]; independent measurements under Intel TDX report 8.9-21.8% [73]

The TEE lane is what makes the protocol adoptable by an organization, and it is affordable: single-digit percentage overhead is the only confidentiality primitive in this space with that property.

The SANITISABLE lane must be described honestly. Against an undefended model on a legal-text corpus, PII extraction reaches ~23% recall and ~30% precision, and PII *inference* from 100 candidates reaches 70%, 50% and 28% on three corpora. Differential privacy at $\epsilon=8$ reduces extraction recall to about 3% — but not to zero [74], and differentially private generation measurably degrades language quality [75]. Sanitization reduces risk; it does not eliminate it, and the interface should say so rather than bury it.

13.5 Dynamic privacy tiers and trusted swarms

The lanes of §13.4 classify *work*. They say nothing about the *population of machines* that work is permitted to reach. A second axis is available and costs almost nothing to add: the topology itself can be tiered, so that the routing decision is a pair — which lane, and which mesh.

Classification, and where it runs. Every request passes a privacy classifier before planning, in two modes. The first is a **manual hard flag** — `--privacy=trusted`, `--privacy=local` — deterministic, declared by the user, and never overridden by the automatic path. The second is **automatic triage**: a small local model performs named-entity recognition over the prompt and raises the tier when it detects entities of regulated classes.

The essential property is that this classifier runs **entirely on the client**. A privacy classifier that consults the network to decide whether the prompt is private has already disclosed the prompt. It is also deliberately recall-oriented: it must over-classify, because the cost of routing a public prompt to a trusted swarm is some throughput, while the cost of the converse is the failure the tier exists to prevent.

The three tiers. Tier 1, the global untrusted mesh, is the default and is the network this document has described: open volunteer nodes, PUBLIC and SANITISABLE lanes, the full verification stack, and redundancy at the k derived in §7.4.

Tier 2, the trusted swarm, is a permissioned sub-mesh. Membership is a cryptographic whitelist of node public keys under an operator-controlled registry; every link carries mutual TLS; the typical deployment is a corporate LAN, a campus network or a VPN overlay. The protocol is unchanged — a trusted swarm is the same protocol over a restricted population, not a second protocol — and that constraint is deliberate. Round-trip times inside such a swarm collapse from tens or hundreds of milliseconds to well under one, which means the bandwidth asymmetry of §1.2 is locally suspended. It would be technically possible to run a finer-grained partition inside the firewall. Swarmbly does not, because a deployment that behaves one way inside the perimeter and another outside it is two systems to implement, verify and reason about. What the low latency buys instead is headroom in the context budget: **larger L , more generous F , higher ρ** , and therefore better coherence at the same wall-clock — an improvement obtained by spending the same design parameter rather than by introducing a new mechanism.

Tier 3, pure local execution, is selected by `--privacy=local` and means what it says: no packet leaves the machine. It is the only tier in which Swarmbly makes an unconditional confidentiality claim, and it can make it precisely because there is no network to make it about.

Redundancy in a trusted swarm. Version 1.4 already observed here that k was serving two purposes at once and that a whitelist removes one but not the other. Section 12.4 generalises that observation to three purposes, and this section is now its particular case. A trusted swarm **MAY** set $k_{\text{verif}} = 1$, because the adversarial defence is already resolved at the identity layer. It **may not** set $k_{\text{epist}} = 1$ without losing the divergence map, and when it does, the response metadata records that no map was produced and the client surfaces that absence explicitly rather than presenting an empty map as agreement. k_{avail} is bounded by the coverage equation: with $c = 1$ there is no margin against loss at all, so the design requires $k_{\text{avail}} \geq 2$ whenever the measured intra-swarm loss rate exceeds the tolerance ε , LAN reliability notwithstanding.

Why the tier matters beyond engineering. Data-protection regimes are written around identifiable, contractually bound processors: the processor relationships of the GDPR, business-associate agreements under HIPAA, and their equivalents all presuppose an entity that can be named, audited and held liable. An anonymous volunteer cannot be a processor under any of them. A whitelisted, mutually authenticated node inside an operator's own registry can. Tier 2 is therefore not a performance feature; it is the construction under which this architecture becomes lawful in settings where Tier 1 is not.

And what it does not do. A trusted swarm relocates trust rather than eliminating it. Whoever controls the whitelist controls the swarm, which makes registry governance a security-critical function; and a compromised member inside the perimeter is *more* dangerous than an untrusted node outside it — precisely because the redundancy that would have caught it may have been reduced. Mutual TLS authenticates the channel and the identity; it says nothing about whether the model behind that identity is the declared one. For that reason the locality-sensitive commitment of §13.3 remains **REQUIRED** inside a trusted swarm even where audit sampling and majority vote are relaxed.

13.6 Sybil resistance: a limitation, declared

Without a trusted identity authority, a single adversary can present arbitrarily many distinct identities, defeating **any** scheme based on redundancy or majority vote [76]. Reputation systems do not escape this: the canonical P2P algorithm requires a set of *pre-trusted* peers to be Sybil-resistant, which reintroduces the anchor it was meant to remove [77].

That this is not merely theoretical is visible in the flagship volunteer-computing deployment, where 41.4% of hosts belonged to single-host users, 44.2% to users with 2–10 hosts, and the largest single user operated 2,987 hosts [47] — extreme concentration, by a benign participant with no incentive to conceal it.

Swarmbly is therefore not Sybil-resistant in the strong sense, and the protocol says so. It adopts layered trust: accumulated reputation, a cost to enter the registry, sampled audit with economic penalty, and a set of anchor nodes operated by the foundation for cold start.

14. Economics, governance and sustainability

14.1 Credits, not tokens

Earlier development proposed a 15% founder premine and a 0.5% protocol fee on each micropayment. Both are withdrawn on regulatory and narrative grounds. The Swiss framework classifies tokens as payment, utility or asset, with a two-condition test to escape securities classification [78]; a premixed, transferable instrument with an expectation of appreciation is the archetype that triggers it. European exemptions are narrow — €1,000,000 over twelve months, or 150 persons per member state, with service-provider rules in force since 30 December 2024 [79].

The design that stays outside both regimes is deliberately unexciting: credits that are non-transferable between accounts, earned by processing and spent by requesting; no presale and no premine; immediate utility from day one; expiring balances to discourage hoarding; and fiat conversion in one direction only — enterprises buy capacity through the commercial arm, volunteers do not sell credits.

14.2 Licence and governance

The protocol implementation is **AGPL-3.0-or-later**. Its clause 13 closes the network-use gap that GPL leaves open [80]. Three qualifications: the obligation attaches to the Program and its modifications rather than to a surrounding proprietary stack; the licence covers software, not the protocol, so a clean-room reimplement is lawful; and its practical force is deterrence rather than litigation.

The strongest empirical guidance comes from the recent relicensing wave: of the four best-known projects that hardened their licences, three produced a successful independent fork (OpenSearch, Valkey, OpenTofu), and two of the four subsequently reverted to AGPL [81, 82]. Starting at AGPL and staying there is the position that history supports.

Two structural decisions follow. Dual licensing is rejected — it requires an entity that can sell proprietary exceptions, which is incompatible with a foundation whose mandate is openness; revenue comes from managed service instead. And **trademark, not copyright, is the operative control lever**. Contribution is by DCO sign-off, not CLA, because copyright assignment creates friction precisely with the community this project needs.

On structure: a Swiss *Verein* can be constituted quickly and without minimum capital, which is sufficient to hold rights and receive grants; a *Stiftung* is the right instrument later. The

claim that a foundation is "unacquirable and unsilenceable" is overstated in any case — foundations are captured through boards, donor dependence and control of repositories and marks. Independence is a practice, not a legal form.

14.3 Sustainability, unclaimed

Data centres consumed 415 TWh in 2024, about 1.5% of world electricity, with projections to 945 TWh by 2030 [83]. US hyperscale facilities draw from grids measured at 545 gCO₂/kWh against a 370 g national average [84]. Those figures support the *motivation*.

They do not support a claim of net benefit, and I do not make one. Energy per token varies by nearly three orders of magnitude across configurations, and datacenter accelerators achieve the lowest energy per token in the large majority of scenarios; idle draw of 12–59 W is paid in full by a node that is available but unused [85]. Global PUE is 1.54, but hyperscalers operate at 1.09–1.15 against a home's effective ~1.0 — a margin of 9–15%, not an order of magnitude [86].

The commitment I make instead is procedural: adopt the Software Carbon Intensity standard — $SCI = ((E \times I) + M) / R$, ISO/IEC 21031:2024, which explicitly excludes offsets [87] — instrument nodes and client, and **publish the result whatever it shows**. The defensible motivating argument is the embodied-carbon one: extending the service life of hardware that already exists avoids new manufacture.

On induced demand. Making a resource cheaper usually increases its total consumption rather than displacing existing use — Jevons' paradox — and a reviewer at a climate fund will raise it. The claim Swarmbly makes is not that this demand disappears, but that **it is absorbed by hardware that has already been manufactured**. Two conditions bound the argument and both are stated: it holds only while **spare capacity exists**, and it holds only for the fraction of traffic served by genuinely pre-existing volunteer hardware, which explicitly **excludes the anchor nodes** of the bootstrap period.

14.4 Bootstrap: anchor nodes, declared

A network whose supply and demand must arrive simultaneously does not start on its own. Swarmbly's bootstrap subsidises supply: the foundation operates rented capacity so that service is fast and stable from the first day, and retires it as community supply grows.

This creates an integrity exposure that the protocol handles by disclosure. During that period a portion of the "volunteer swarm" is rented datacenter hardware, and for that portion **the embodied-carbon argument does not apply and the decentralization claim is only partially true**. The commitments are therefore explicit: such nodes are named **anchor nodes operated by the Foundation** and are labelled as such in the registry; the public dashboard reports, in real time, **the share of traffic served by anchor nodes versus community nodes**; and the Foundation publishes a target trajectory for that share and reports against it.

15. Evaluation

15.1 Task categories

Fit requires five simultaneous attributes: decomposable into genuinely independent sub-tasks; latency-tolerant; token-intensive; weak inter-fragment dependencies; verifiable or non-sensitive content.

Category	Fit	Reasoning
Bulk document processing	High	Embarrassingly parallel, no dependencies, latency-tolerant
Synthetic data generation, labelling	High	Independent per sample; filter-verifiable; large volume
Code-migration sweeps	High	Independent per file; verifiable by compiling and running tests
Evaluation and judging at scale	High	Independent; aggregation is a vote
RAG over large corpora	Moderate	Map stage parallel; but fewer, larger units beat many small ones [52]
Long structured reports	Moderate	Decomposable by section — and precisely where SoT degrades [12]
Multi-hop mathematical reasoning	Poor	Result-dependencies [51]
Code with shared mutable state	Poor	The canonical failure through conflicting implicit decisions [33]
Interactive low-latency chat	Very poor	Lossless single-node methods already dominate [13, 14, 15]

[v0.3] Under §6.8, this table admits a reading v1.4 did not have: the poor-fit rows are the **high-δ** rows, and the high-fit rows are the low-δ ones. That is not a coincidence of wording — it is what the $D(\rho, \delta)$ surface predicts, and it turns the table into something checkable rather than a list of intuitions.

15.2 The abandonment criterion

Go/no-go. There must exist a ρ at which coherence degradation is **below 5% relative to monolithic generation, in at least one task category**. If no such ρ exists, the architecture is not viable for generative assembly, and the project should either stop or restrict itself to workloads with no seam to break — classification, extraction, labelling.

How the criterion is applied, and why the form matters. "Below 5%" is a claim about a quantity estimated from a finite corpus, so the criterion is discharged against an **interval, not a point estimate**: the upper bound of a 95% bootstrap interval, clustered by prompt, must fall below 5% in the named cell. That clause is part of the original pre-registration and not a later addition — which is the only reason it is worth anything, since a criterion tightened after the data arrive proves nothing and one loosened after they arrive proves less.

[v0.3] And this version adds a second baseline, from the Zhang et al. criticism (§2.4): additional compute must also be justified against **a single agent with self-consistency**, not only against the naive monolithic baseline. A multi-agent architecture that only beats single-pass monolithic generation is comparing itself against the wrong rival.

The criterion is not measured — and the cleanest measurement is a cost. The aggregate tax is confounded with output length in **both directions** (full budget: the fragmented arm writes **1.39×** the monolithic and scores more; divided budget: **0.43×** and scores less), and truncating to a fixed reading budget **T** flips the sign of the tax with **T** (**−15.66 %** at **T=40** to **+7.69 %** at **T=120, T13**). The position-of-first-mention instrument (T13b, SWIP-0001), corrected so that an omitted key is charged at the end of the text (worst case), also does not decide: displacement **+0.0036** of the arm's own length, 95 % CI **[−0.0948, +0.0994]** — a null with an interval $\sim 5\times$ wider than the threshold — and the two corpus halves disagree in sign, so the instrument **refuses** (the worst-case imputation also recouples the metric to length: $\rho = -0.452$, over 423 keys of which the fragmented arm omits **217** against the monolithic's **82**). The measurement that does survive is economic: under the equalised budget, fragmenting **costs in all five families** (over the 95 criterion cells, T09R:

means **+6.2 %** to **+22.1 %**, medians **+4.3 %** to **+29.4 %**; aggregate **+11.17 %**, 95 % CI **[+4.80, +17.34]**, T09R), and the measured per-class policy is **not to fragment (+0.0 %** against **+14.8 %** for always fragmenting). That is the result this section reports first, because it is the one the data support.

15.3 What was measured, and what was withdrawn

The coherence tax: criterion NOT MET. table_summary, $\rho = 3.5$, $N = 2$, $k = 1$, on 16 held-out prompts:

Coherence tax	+2.30%
95% CI (clustered by prompt)	[−2.05%, +7.49%]
Criterion	upper bound below 5%
Verdict	NOT MET — 2.49 points short on the upper bound
Median prompt effect	exactly 0.00%
Prompts at or below zero	11 of 16 (6 negative, 5 exactly zero)
Control, $N = 8$, $k = 1$ (required to fail)	+16.23%, CI [+11.33%, +20.28%] — behaved

The point estimate clears the threshold and the interval does not. The criterion was written against the upper bound for exactly this case, and it is not being rewritten now that the case has arisen.

The distribution is the more informative result, and the mean is the wrong statistic for it. The median prompt loses nothing, eleven of sixteen are at or below zero, and the mean is not merely pulled by the positive prompts — it is *manufactured* by two of them, which sum to **140%** of the total; remove them and the mean over the remaining fourteen is **−1.06%**. The honest description is not "fragmentation costs 2.3%" but **on 11 of 16 table-summarisation prompts, splitting the work in two was free, and on two of them it was expensive.**

Two withdrawals from v1.4 that still stand. The tax curve falling in ρ is withdrawn: the ρ axis never moved, because 13 of 96 cells sat above their packing floor, and the coherence metric was not arm-neutral (+46.7% of apparent tax on text that never changed). And the confidence map's reliability claim is withdrawn: $r = -0.030$ over 597 units against a saturated judge, and then common odds ratios of 3.47, 0.26 and 1.24 against an answer key. §10.5.4 explains why M1 does not lift that withdrawal.

15.4 The L curve: the instrument works, the corpus does not

An experiment was built and run to measure the quality curve against L — the parameter Section 5 makes central. **It could not answer the question, and the reason is informative.**

The corpus was generated with 72 documents derived from the product of $L \in \{5,10,20,40\}$ and $N \in \{2,4,8\}$, with 12 documents per size. The monolithic arm answers **1 of 72** global questions, even on the smallest material. That is not a result about fragmentation: it is a floor. With the baseline on the floor there is no difference to measure.

Verifying the instrument is what saved the run. Before concluding anything, the grading was checked: a hand-built perfect answer scores 100% on all 72 documents. The grader is not the problem. Five model families were then probed on the same material: **4 of 5 do the local lookups at ~0.84**, and global questions come out at **4 of 60** across all of them. A hypothesis that the contract wrapper was strangling the global answer was tested with an A/B and **was refuted by its own measurement:** 4/60 both ways, delta 0.000.

Conclusion, and it is about corpus design rather than architecture: the corpus needs a *global* question this model pool can answer on small material. Until there is one, the L curve is not measurable, and the final half of the corpus — still unused — **must not be run**, because spending the held-out split against an instrument that does not discriminate destroys the only reserve left.

§15.4 update (corpus rebuilt and admitted; the first curve run was invalid and was withdrawn). The corpus was rebuilt with k-ary global questions (arity ≤ 3) — prompts/lcurve_v2.json — and passed its admission gate: the monolithic arm clears the 0.50 floor with llama3.2:3b at **61.1 %** (chance baseline **0.182**; the other four families do not clear it). The first fragmented run over the declared cells (N, L) **was withdrawn**: each fragment copied its rows verbatim, the concatenation reconstructed the original document, and the model answered the same text as the monolithic arm — the 48 cells coincided with their monolithic counterpart one for one (48/48), so the run measured nothing about fragmentation, and L was confounded with document size. The redesigned run (each fragment *answers* per-row values and its own partial sum; the assembler combines deterministically, run_lcurve_v2.py) does measure something about fragmentation, but **its assembler computes only 3 of the 6 global question forms**: pair_sum, triple_argmax and pair_warehouse — 32 of the 72 dev globals — are scored as failures by construction. Over the run's cells the assembler covers 79 of 144 globals (54.9 %), and that is the fragmented arm's ceiling: at L=5 and L=10 it reaches it exactly. Compared against the monolithic arm on the same document, question by question (T08R2):

L	computable	fragmented (computable)	monolithic (computable)	fragmented (all)	monolithic (all)
5	23/36	100.0 %	73.9 %	63.9 %	77.8 %
10	20/36	100.0 %	75.0 %	55.6 %	69.4 %
20	19/36	73.7 %	63.2 %	38.9 %	58.3 %
40	17/36	52.9 %	47.1 %	25.0 %	44.4 %

On the questions the assembler can compute, the fragmented arm **beats** the monolithic arm at every L: paired difference **+18.9** points, 95 % CI **[+1.4, +37.5]**. Over all globals the difference is **-16.7** points **[-27.8, -4.6]**, but that figure measures the assembler's coverage and not fragmentation, so the architectural verdict **refuses**. Two caveats on the advantage: part of it is by design — the fragmented arm aggregates with exact code while the monolithic arm does its own arithmetic, which is precisely what the protocol proposes — and the sample is small (79 questions, one family). Three further limits: the slope in L is not measured (each L uses its own band of sizes; across the 15 documents with two or more L there are 11 ties, 3 favouring the smaller L and 1 the larger); value fidelity is not measured either (the record counts answered rows, not correct ones); and the partial sums are not verifiable without the raw answers. run_lcurve_v3.py implements all six forms, runs every L on the same documents, and stores per-row fidelity and raw answers. On the v2 run alone, this section **claims no parity, cost or win** on the L curve.

The v3 run exists (data/lcurve_v3_runs.jsonl): an assembler with all six forms, L = 5, 10, 20 and 40 on the same 8 documents of 80 and 160 rows, per-row fidelity and raw answers; the monolithic arm was re-run under the same conditions and came out identical, byte for byte, to the admission run. Result (T08R3):

L	fragmented	monolithic	extraction fidelity
5	24/24 = 100.0 %	12/24 = 50.0 %	960/960
10	24/24 = 100.0 %	12/24 = 50.0 %	958/960

L	fragmented	monolithic	extraction fidelity
20	24/24 = 100.0 %	12/24 = 50.0 %	960/960
40	21/24 = 87.5 %	12/24 = 50.0 %	923/960

Paired difference **+46.9** points, 95 % CI **[+33.3, +59.4]**. Within document the slope in L is flat up to L=20 and drops at L=40: 2 documents get worse, none gets better and 6 tie. The raw answers are auditable: the fragments extract almost without error, but sum their own block correctly only 1 time in 360.

And the control that separates the two effects has run (T08R4): the same extraction over the whole document (N=1), with the same reader and the same assembler, on the same 8 documents. In 3 of them the model answered without numbering and was read by position, re-graded from the raw answers.

rows	monolithic	N=1 + code	N=1 fidelity	fragments + code	effect of aggregating by code	effect of splitting
80	58.3 %	100.0 %	314/320	100.0 %	+41.7	+0.0
160	41.7 %	16.7 %	158/640	93.8 %	-25.0	+77.1

Across the 8 documents, the effect of splitting is **+38.5** points, 95 % CI **[+12.5, +68.8]**, and that of aggregating by code **+8.3** **[-16.7, +33.3]**. But the average hides what matters: **these are two different mechanisms, and each dominates in its own regime**. At 80 rows, extracting the whole document in a single call is already near-perfect, and all of the gain comes from taking the arithmetic away from the model; splitting adds nothing. At 160 rows, single-call extraction collapses — the model loses the format and loops: 1,140 lines with barely a hundred distinct values — and only splitting keeps it near 100 %. It is the first direct measurement of the §7.5 thesis that the worker size that suffices depends on the budget it is asked to carry: a 3B node extracts well up to some task size and falls off a cliff above it, and fragmenting is what keeps each task below that threshold.

Limits, declared. These are 8 documents, one family and two sizes, so the threshold lies between 80 and 160 rows and has not been located. What collapses may be the input length, the output length (321 answers per call) or both; this design does not separate them. The monolithic grader is lenient, which works in its favour. And the conclusion does not depend on reading by position: without it, the effect of splitting is larger (+42.7).

15.5 Composition: the sample size required

The declared cell of the composition experiment gave a mean of **-0.35**, with a between-cluster standard deviation of **20.36** and a standard error of **4.16**, for a CI of **[-8.49, +7.80]** against a threshold of 5.0. The interval contains both the threshold and zero: it decides nothing.

The sample-size calculation, cross-checked against the measured standard error, says **60 prompts minimum and 72 with margin**. And there is a useful negative result along the way: **repeats buy nothing**, because the pipeline is deterministic at temperature 0. More runs of the same prompt do not reduce the between-prompt variance, which is what dominates.

Two composition measurements are conclusive and do support decisions in this version. Mechanical enforcement of `term_once` at the assembler raised compliance from **6/24 to 18/24**, above the monolithic arm's 13/24 — it is the evidence that resolving a constraint in the system rather than in the prompt works, and it is the baseline M2 must beat (§11.3). And `no_repeated_ngram` stayed at **4/12 against 11/12** for the monolithic arm under every context-allocation policy tested — it is the irreducible class, and §6.4 explains why no allocation can reach it.

15.6 What each model predicts, and how to kill it

The five models of this version are falsifiable. Stating it here is what separates a major version from a manifesto, and it is the condition under which this document can be published.

M1 — Alignment with learned semantic substitution (§10.5). *Predicts:* a map built by alignment with learned cost separates correct from incorrect better than one built by match counting, and better than a scalar per answer, **in a non-saturated regime**. *Killed if:* with the correct instrument and in a regime where the models genuinely disagree, per-unit agreement does not beat chance. Then localisation creates no signal and the map loses its reliability claim, though it retains its ability to report divergence.

M2 — Cardinality resolved in the representation (§11.3). *Predicts:* assigning uniqueness in the plan makes `no_repeated_ngram` and `term_once` inviolable by construction, not merely detectable. The violation rate should fall to zero, not improve. *Killed if:* assignment in the plan does not reduce violations below 18/24, which is what mechanical enforcement already achieves.

M3 — Per-level triage gate (§10.7). *Predicts:* the blast radius of a defective fragment stays contained in that fragment. An incident of the "42 of 60 packets contaminated" type becomes impossible. *Killed if:* the gate rejects so much legitimate work that the retry cost exceeds the damage it prevents.

M4 — ρ as indirect rate-distortion with a second axis δ (§6.7-6.8). *Predicts:* at constant ρ , varying the dependency density of the cut moves the distortion. And there exists a region where no ρ reaches an acceptable distortion. *Killed if:* distortion turns out to be a function of ρ alone, with δ having no measurable effect. Then the second axis is decoration and the framing adds nothing over the empirical derivation of §6.5.

M5 — Threshold over-dispatch, and k split into three (§12.3-12.4). *Predicts:* separating availability, verification and epistemic redundancy allows the total cost to be lowered while keeping all three guarantees, because today a single parameter pays for all three at the price of the most expensive. *Killed if:* the three turn out to be so coupled in practice that separating them saves nothing.

Status after the reference campaign (§15.9).

Model	Verdict	On what
M1	no instrument	the semantic aligner is built and verified on the bench (T10), but the real corpus did not produce the non-saturated regime M1 needs (T10R, blocked)
M2	survives, corrected	23/35 by node obedience; 35/35 with mechanical enforcement. The phrase "zero by construction" is withdrawn (§11.3)
M3	survives	contaminated packets do not pass the gate; the 42/60 incident does not reproduce (T02R)
M4	falsified	in two independent instruments and with a controlled experiment at constant ρ (T04R, T07R)
M5	unmeasured	it remains a conceptual separation; nothing in this campaign touches it

The order in which to attack them, by increasing cost and by what each one unlocks:

1. **M5** needs no measurement: it is a conceptual separation of a parameter that already exists. Implement and observe.
2. **M3** is cheap and its prediction is binary: the incident happens or it does not.
3. **M2** is measured over the composition corpus that already exists, against a baseline already measured (18/24).

- 4. **M4** needs the difficulty-calibrated corpus on which the entire current experimental agenda depends — the same one §15.4 says is missing.
- 5. **M1** needs the aligner to be built, and is the most expensive. It is also the one that produces the most distinctive property.

15.7 A note on the instrument, which is the methodological finding

This project has a recurring defect, and naming it is worth more than any of the individual measurements:

A check that claims more than it measured.

An aggregate is reported without asking where it comes from. It appeared four times in a single working day, and **each time inside code written to catch the previous occurrence**. The cases: a verdict reading a key the bootstrap function did not return, which would therefore have said "not met" on any data at all; a probe that silently degraded to one model family and concluded about five; a difficulty diagnosis derived from a single model and presented as a property of the corpus; and a threshold that fired on the effect of a single cell.

The defence that worked was not more care. It was **making the check refuse**: deriving the family list from the runner itself and refusing below two; withdrawing a pooled conclusion if removing the largest contributor undoes it; cross-checking the sample-size plan against the measured standard error and refusing if they differ by more than 25%. An instrument that can refuse is worth more than one that is right more often.

Four thresholds in this document exist for that reason and are declared: a minimum of **20 clusters before a verdict is issued**; a **baseline floor of 0.20** — added *after* seeing the data and labelled as such in the pre-registration, because a declared amendment counts and a silent one does not; and the two control tolerances of the previous paragraph.

15.9 The reference campaign: 473 runs over five families

Everything earlier in §15 was measured with the project's harness over the project's corpus. This section reports something different: a **reference implementation built separately** (`swarmby_ref/`), run against **five local model families** — `llama3.2:3b`, `qwen2.5:3b`, `gemma2:2b`, `phi3.5:3.8b`, `granite3.1-dense:2b` — over a corpus of 22 tasks, and a **falsification harness** (`swarmby_validation/`) that judges the five models against the resulting record. The record is a JSONL of **473 runs** with an anchored digest; all three artifacts are published with this document, and every figure below is recomputed by running them.

What has to be said first is what died.

Test	Verdict	What it measures
T0R	✓	the router decides decomposable/not against corpus truth: 22/22
T02R	✓	M3: contaminated packets do not pass the gate
T03R	✓	M2: 23/35 by node obedience, 35/35 after mechanical enforcement
T04R	✗	M4: Spearman(δ , tax) = -0.15 over 127 intra-category cells
T05R	✓	the ρ accounting predicts packet size to within 7.5 %
T06R	✓	the grader scores the perfect answer 21/21 ; 100 of 105 monolithic runs clear the floor
T07R	✗	M4, controlled experiment: δ rises in 32/32 pairs and the tax worsens in only 15/32
T08R	✗	L curve: the predicted band (factor 3-6 $\times$) does not appear
T08R2	⦿	L curve on the admitted corpus: the assembler covers 54.9 % of the globals, so the verdict refuses; on the computable ones, +18.9 points in favour of the fragmented arm

Test	Verdict	What it measures
T08R3	✓	L curve v3: fragments + code +46.9 points over the monolithic arm; mixes splitting with aggregating by code
T08R4	✓	N=1 control: splitting +38.5 points; at 80 rows the gain is from aggregating by code, at 160 rows from splitting
T09R	⊙	abandonment criterion: refused , confounded with output length
T10R	■	M1: no non-saturated regime in this corpus
T0RR / T0RR2	✗	per-cell routability: AUC 0.57 and 0.57
T0LR	✓	L* varies by family — \$5.7 measured for the first time
T13	⊙	the truncated tax flips sign with the reading budget T — the length coupling is structural, and the truncation instrument is refused
T13b	⊙	position tax, corrected for omissions (SWIP-0001): displacement +0.0036 [−0.0948 , +0.0994], halves disagree in sign — the criterion is not measured

M4 is falsified, and the experiment that kills it is the one the strategy declared decisive. T07R cuts the **same prompt** twice at the **same L**: once at weak-coupling interfaces (P9) and once maximising the coupling that crosses the cut. δ rises in all 32 pairs and ρ stays equal to within 5 %, so the only axis that moved is δ . M4 predicts that the strong cut worsens distortion in **every** pair; it worsens in **19 of 32**, which is indistinguishable from a coin. The observational measurement converges: Spearman(δ , tax) = -0.15 over 127 intra-category cells, with the sign opposite to the prediction. **The second axis is not conceptual decoration: it is a prediction that was checked and did not hold.**

The per-cell router is refuted, and this changes the design, not only the scoreboard. A classifier with leave-one-task-out validation over 80 operational cells reaches AUC 0.57 with δ and reputation and 0.61 with neither. Adding pre-dispatch capability probes brings it to 0.57. What does separate is the **node class**: mean tax per family ranges from +8.4 % to +26.5 %, and a policy that routes by class obtains +0.0 % against +14.8 % for always fragmenting. The consequence is written into §10.1: the router decides *whether* to fragment, the node profile decides *who*, and nobody predicts the cost of a particular cell.

M2 survives with a correction worth more than the confirmation. Real nodes do not obey unique_here: 6 of 35 terms appear exactly once, with 29 repetitions. But **0 omissions** — and the assembler's mechanical enforcement takes the result to 35/35. The failure mode that assignment in the plan removes is the irreparable one; the reparable one is still removed by the assembler. §11.3 rewrites M2 in those terms and withdraws "zero by construction".

The abandonment criterion has neither been met nor failed: it has not been measured. The aggregate gives +11.17 % with a 95 % CI of [+4.80, +17.34] over 130 cells, which would appear to settle the criterion in favour. It cannot be claimed, because the fragmented arm received more output budget than the monolithic one: each fragment inherited the full max_tokens of the whole prompt, so a plan of N fragments had N times the output available. Three independent statistics confirm this suffices to explain the advantage: the statistic with no denominator gives **+0.595**, the artifact floor — the same calculation with the tax permuted — gives only **+0.125**, and the direct test within each task gives a median of **+0.571**, positive in 10 of 18 tasks.

Stratifying by length, the aggregate comes apart where it should:

Stratum	n	tax	95 % CI	meets the criterion?
all cells	123	−32.5 %	[−54.7, −12.0]	yes

Stratum	n	tax	95 % CI	meets the criterion?
comparable length (0.8-1.25×)	31	−21.3 %	[−42.0, −1.0]	yes
the fragmented arm does NOT write more (≤ 1.0×)	41	−6.3 %	[−22.3, +14.1]	no
the fragmented arm writes more (> 1.25×)	67	−54.4 %	[−77.1, −24.6]	yes

The row that matters is the third: when the fragmented arm does **not** write more, the advantage falls to −6.3 % and the interval crosses zero. This is a post-hoc stratification and not an experiment, so it settles nothing on its own — but it is exactly the pattern the confound predicts, and that is why the verdict is **refuse** rather than "met with caveats".

The cause was located in the code, not conjectured: `benchmark.py` gave each fragment the whole task's `max_out_tokens`. The correction is applied and exposed as a single command — `run_benchmark.py --matched` — which divides that budget among the fragments and records the cells under a suffix of their own so that the two measurements coexist. The fields `output_words` and `length_ratio` are now first-class fields of the record, so that a future analysis cannot repeat the error silently.

What this campaign does not touch. T07R, T04R, T05R, T06R, T02R, T0R, T0LR and T10R compare arms at equivalent budget or do not depend on the aggregate tax, so the confound does not reach them. The verdicts on M2, M3 and M4 stand whole.

And a fifth appearance of the §15.7 defect, this time in the analysis of this very section. Exploring the data, an attractive interaction appeared: weak nodes seemed to gain from fragmentation and strong ones to lose. Before confirming it, a pre-registration was written declaring the suspected defect — $\text{tax} = 1 - \text{frag}/\text{mono}$ carries `mono` in the denominator, so correlating it with the same cell's `mono` produces association *by construction* — and the design that breaks it: estimating node strength from that model's results on the **other** tasks. With the independent predictor, $\rho = -0.078$ ($p = 0.192$); and a simulation where the two scores are drawn independently returns **+0.684**, against the **+0.675** observed with the coupled statistic. The formula manufactures the whole association. **The finding was withdrawn**, and the test that killed it is published with the harness (T12) so that the withdrawal is reproducible rather than a footnote.

15.8 Metrics

Metric	Definition	Target	Status
Coherence tax	Δ seam-free sentence fraction vs monolithic	upper bound of the 95% CI below 5%	NOT MET on the table corpus (§15.3); REFUSED in the reference campaign owing to the length confound (§15.9)
Operating ρ	Input tokens per prompt token	distance to $(L+2F)/L$	Re-derived in §6.5; v1.4's absolute " <2.0 " target is withdrawn as neither attainable nor meaningful
Quality curve in L	Quality vs fragment size	existence of $L_{\min}$ and of a band	measured per family : L^* varies across families, but the predicted wide band does not appear (§15.9, L20)
Effect of δ at constant ρ	Distortion vs dependency density	measurable effect	measured and null : δ rises in 32/32 pairs at equal ρ and the tax worsens in 15/32 (§15.9)

Metric	Definition	Target	Status
Cardinality violations	over-compression and over-expansion, separately	over-compression zero by construction; over-expansion zero after enforcement	over-compression 0/35; over-expansion 29/35 raw and 0/35 after enforcement (§15.9)
Triage containment	blast radius of a defective fragment	one fragment	contained: all 3 contaminated packets are rejected (§15.9)
Effective speedup	vs monolithic, same model	>1.5×	not measured
Speedup vs honest baseline	vs speculative decoding and vs self-consistency	reported even when <1	not measured
p95 latency under churn	$p=0.10$, $N=8$	<2× failure-free	not measured
Dishonest-node detection	injected adversaries caught	>95%	not measured
Verification overhead	extra cost per fragment	<5%	not measured
SCI	gCO ₂ e per functional unit	published and compared	not measured

16. Limitations and negative results

Stated plainly and at length. A specification whose failure modes are documented can be improved by people who did not write it; one that hides them can only be discovered to be wrong.

L1 — Quality loss from independent generation is theoretical, not incidental. Parallel generation assumes conditional independence, and quality degrades in proportion to the strength of the real dependencies [20]. At equal compute budget, decomposition is a lossy channel [21]. This cannot be prompt-engineered away; it can only be routed around, which is why §10.1 exists. **[v0.3]** And §6.8 gives it a name and an axis: it is δ .

L2 — Coherence is the axis that breaks, and aggregate scores hide it. SoT improves relevance and diversity while degrading coherence and immersion [12]. Hierarchically merged text exhibits eight recurring coherence error classes [54].

L3 — The assembler operates in a regime known to be unreliable. Model reliability degrades with input length across all models tested; one distractor hurts and four compound; and models perform *better* on shuffled contexts than on logically coherent ones [88]. Positional bias adds a U-shaped curve in which middle fragments are systematically underweighted [89].

L4 — The context limit is relocated, not eliminated. It moves to the client, which is the weakest node in the system.

L5 — An 8B orchestrator may be inadequate. This is H3, and a negative result would require a larger client-side requirement, which narrows the addressable user base.

L6 — No strong Sybil resistance. See §13.6.

L7 — Fragmentation is not encryption. See §13.1.

L8 — Environmental benefit is unproven. See §14.3.

L9 — The genomic analogy is vocabulary plus instruments, not an inheritance. [v0.3]

This limitation changes shape from v1.4. It is no longer "it is only a naming convention": §3 gives a criterion, and five homologies pass it bringing concrete inequalities and numbers. But it remains true that **no genome-assembly algorithm runs inside Swarmbly**, and a reviewer from bioinformatics should read §5, §6.4 and §11 as transfers of *instrument*, not of implementation.

L10 — The dominant risk is not technical. Volunteer computing has been in decline for two decades: early projects attracted on the order of a million volunteers, and the user base has since shrunk to roughly two hundred thousand [47]. Swarmbly must explain what makes its incentive loop different, and "network credits" is not by itself an answer. This is, in my assessment, more likely to end the project than any algorithmic limitation.

L11 — Multi-agent systems fail in characterized ways. The MAST taxonomy attributes 47.9% of failures to system design, 32.2% to inter-agent misalignment and 20.0% to task verification, with step repetition (15.7%) and specification disobedience (11.8%) the most common individual modes [90]. Swarmbly is a multi-agent system and should expect this distribution. **[v0.3]** §2.4 maps the three categories against the three areas this version addresses; that coincidence is a convergence, not an immunity.

L12 — A trusted swarm relocates trust; it does not remove it. See §13.5.

L13 — The confidence map's reliability claim remains withdrawn. Measured against a peer-class judge, per-unit agreement did not predict judged acceptability ($r = -0.030$ over 597 units), and $k > 1$ cost 17 to 20 points of coherence on the same run. The ground-truth experiment was run three times and returned odds ratios of 3.47, 0.26 and 1.24. The mechanism stays disclosed and specified, and reporting *where* replicas diverged is still a real capability; asserting that convergence indicates correctness is not. **[v0.3]** M1 proposes a different instrument with its own death condition (§15.6); proposing it does not lift this withdrawal, and nothing in this version should be read as if it did.

L14 — The second coherence instrument does not function on short answers. The entity grid returned monolithic baselines between 0.000 and 0.114. This document has one working coherence instrument, not two.

[v0.3] L15 — The L curve is not measurable with the current corpus. The monolithic arm answers 1 of 72 global questions. The grader is verified and the five families do local lookups at ~ 0.84 , so the defect is in corpus design rather than in the instrument or the models — but the practical effect is that $\mathcal{L}$, the parameter Section 5 makes central, **has no measured curve yet**. The held-out split must not be spent until an answerable global question exists (§15.4).

[v0.3] L16 — $\mathcal{L}_{\min}$ and the optimal band are predicted, not measured. §5.6 predicts a hard floor and a wide band by analogy with protein domain size, where Pfam and SCOP differ by almost a factor of two over the same material. It is a well-founded prediction, and it is a prediction. Any $\mathcal{L}$ this project uses today is an informed guess.

[v0.3] L17 — M2 leaks structure. The `unique_here` field hands a node global information about the session that v1.4 did not hand it (§13.2). Cardinality correctness and decontextualization are antagonists, just as coherence and decontextualization are, and for the same reason.

[v0.3] L18 — Over-dispatch applies only to interchangeable sub-tasks. The saving in §12.3 is real and it is narrow. If the content of fragment j is needed, no threshold scheme substitutes for it, and believing otherwise is exactly the error that sank the fountain-code hypothesis.

[v0.3] L19 — The rate-distortion bound exists and is not computable. §6.7 inherits the existence of a floor and the qualitative monotonicity of the curve. It does not inherit a number. A reader who takes from §6 the impression that a calculable optimal ρ exists has taken more than the document says.

L20 — The wide band in L is predicted and does not appear. §5.6 predicts a sharp $L_{\min}$ floor and a wide optimal band, by analogy with protein domain size. With curves of more than one point per task, `longform` gives a band of factor **2.7** and `table_outturn` one of factor **1.0** — that is, a single optimal L . The analogy was right that L is a property of the node class (T0LR measures a different L^* per family) and wrong about the width of the band. The mean gain from using each family's L^* instead of a common L is **+0.360** of quality: real, measurable and small.

L21 — The measured tax depends on the output budget, and until it is matched there is no feasibility verdict. This is the most important limitation of this version. All of the §15.9 material on M2, M3 and M4 stands; the claim "fragmenting pays" **has not been measured**. The matched-budget run has been done and gives a cost, not an advantage (T09R); but that aggregate is still confounded with output length in both directions, so the feasibility verdict **refuses** (§15.9).

L22 — The recurring defect of §15.7 appeared a fifth time, inside the analysis that documents it. The defence that worked was not more care: it was pre-registering the suspected defect and the design that breaks it **before** running the analysis. With the coupled statistic and a threshold chosen by eye, the false finding would have come out confirmed with large numbers. A reader should assume this document contains a sixth appearance not yet detected, and the published harness exists so that it is the reader who finds it.

17. Prior-art declaration

Status note. Elements E1-E18 were publicly disclosed in version 1.4 of this document, dated 14 August 2026, and have constituted prior art since then. **Elements E19-E24 are new in this version and constitute prior art from the publication of this document**, not before. The period in which they were kept unpublished was a deliberate decision not to claim before measuring, and its cost was exactly the priority it did not protect in the meantime.

The elements are disclosed with the intent that they enter the public domain for patenting purposes; the author reserves copyright in the text under CC BY 4.0 and licenses the implementation under AGPL-3.0-or-later.

E1. A method for distributed language-model inference in which the unit of distribution is a **semantic sub-task derived from the request**, dispatched once per fragment per session to nodes each executing a complete independent model. (§1.2, §8.2)

E2. A **context budget** S as an explicit protocol parameter jointly governing assembly coherence, fragment verifiability, privacy leakage and required worker capability, with the redundancy ratio ρ as the reported cost measure. (§6)

E3. A **global contract** Γ transmitted with every fragment as the mechanism for cross-fragment consistency, with the entity `table` serving as a shared naming coordinate system. (§9.2)

- E4. A router with asymmetric decision cost** that may decline to fragment, calibrated with F_β where $\beta < 1$. (§10.1)
- E5. A dependency-DAG planner** distinguishing tasks requiring a predecessor's *result* from tasks requiring only its *statement*. (§10.2)
- E6. A packing procedure** in which the global contract is never elided to meet a context budget. (§10.3)
- E7. A select-then-splice assembler** in which multiple candidate fragments are resolved by selection rather than synthesis, with generative bridging invoked only at a seam whose boundary similarity falls below a calibrated threshold. (§10.4)
- E8. Empirical calibration of the seam threshold** from labelled pairs under an asymmetric objective, re-derived per embedding model. (§10.6)
- E9. A coherence audit** returned as part of the protocol response. (§10.8)
- E10. Sensitivity-lane routing** — PUBLIC / SANITISABLE / SENSITIVE — as the confidentiality mechanism. (§13.4)
- E11. A two-layer verification scheme** combining a locality-sensitive commitment over activations bound to a declared node profile, with sampled public audit indistinguishable from real work. (§13.3)
- E12. Diversity-preserving redundant dispatch:** replicas of a critical fragment are assigned to nodes of deliberately *different* model families. (§7.3c, §9.6)
- E13. Hedged dispatch at a per-class p95 latency trigger** with cancellation accounting that distinguishes slow nodes from dishonest ones. (§8.4, §9.6)
- E14. A non-transferable, non-premised, expiring credit** earned by verified fragment processing. (§14.1)
- E15. The coverage/conversion decomposition** $Q \approx V \cdot \prod_i C_i$ as a design instrument for swarm inference. (§7.2, §7.3)
- E16. Consensus by multiple alignment of independently generated replicas**, with a per-unit agreement score and units below threshold surfaced as low-confidence regions. Disclosed as a **mechanism**: its correlation with correctness was measured and did not survive, so no reliability benefit is claimed for it. (§10.5, §15.3)
- E17. A coverage model for semantic assembly** in which the pre-generation plan and contract serve as the reference sequence, packet loss rather than sample placement is the stochastic element, and the redundancy requirement is derived as $c \geq \ln(1/\epsilon)/(1-p)$. (§7.4)
- E18. Dynamic privacy tiering with trusted swarms:** a client-side privacy classifier that never leaves the machine, driving a routing decision on an axis orthogonal to content sensitivity. (§13.5)

New elements in this version — not yet publicly disclosed.

- E19. A homology-admissibility criterion** for transferring mechanisms between scientific domains, consisting of two joint tests: that the homology supplies an applicable procedure, inequality or number (the instrument test), and that it supplies the statement of the pathology that occurs when its condition is violated (the failure-mode test); with explicit rejection of transfers that pass only the first. (§3)

E20. A weak-interface fragmentation rule, in which the fragment boundary is chosen by minimising the dependencies that cross it rather than by dividing into parts of uniform size; with the size L as a declared property of the node class, the fragment count derived as $N = \lceil \text{material}/L \rceil$, and the flank F defined as the maximum of the longest detectable repeatable unit and the longest dependency crossing a boundary — from which the context budget is derived as $p \approx (L+2F)/L + H/(L \cdot s)$. (§5.4, §5.7, §6.5)

E21. A divergence map built by multiple alignment of answers with a semantic substitution matrix learned empirically from sentence pairs verified as interchangeable, with consistency extension via corroboration through third independent answers, and emission of a position-specific profile — replacing embedding-similarity comparison against a threshold. (§10.5)

E22. A method for satisfying global cardinality constraints in the plan: compute before dispatch the set of elements that must appear exactly once in the final output, assign each element to exactly one fragment as a property of the plan, verify cardinality against that set at assembly, and log over-compression and over-expansion separately. (§11.3)

E23. A per-topological-level triage gate in which a carry must pass a mechanical predicate before enabling the next level, a fragment that does not pass is isolated and retried rather than patched in context, one that does not pass after r retries is discarded and regenerated, and consumption may begin before the level is complete if the fragment being consumed is complete and verified. (§10.7)

E24. The separation of the redundancy parameter into three single-purpose parameters — availability, verification and epistemic redundancy — with a distinct mechanism assigned to each (threshold over-dispatch with early termination; credibility-based spot-checking; diverse-family replicas for the divergence map), and with the normative condition that over-dispatch is admissible only among genuinely interchangeable sub-tasks. (§12.3, §12.4)

18. Conclusion

The knowledge required to build language models is public. The capital required to operate them is not, and that asymmetry — not any secret — is what concentrates control over a general-purpose technology. Meanwhile the hardware capable of serving inference sits idle in hundreds of millions of homes and offices, already manufactured, already drawing power.

The obstacle between those two facts is physical and specific: four to five orders of magnitude between datacenter interconnect and consumer links, which every existing peer-to-peer inference design crosses on every generated token. This work's structural claim is that crossing it **once per unit of work instead of once per token** is a different regime rather than an optimization of the same one.

What this version adds to that claim is a theory of the unit of work. Version 1.4 knew that fragmentation was necessary and did not know into what. This version derives it: the fragment is a task domain, it is cut where the coupling is weak, its size is a measured property of the node class, its overlap is the maximum of two observable quantities, and the context budget follows from those rather than being swept. The measured saving from that single derivation is 29% of network compute, and it relaxes no guarantee because the theory says exactly what to measure.

Five falsifiable models accompany that theory, and all five come with their death condition written before the measurement. Two hypotheses entered the review and did not survive —

mate pairs and *fountain codes* — and the diagnosis of why they fell turned out to be more useful than the homologies would have been: **a transfer that brings a mechanism without its pathology is importing the conclusion without the work**. That is this version's methodological finding, and it is what Section 3 turns into a criterion.

The case against is equally specific and is in Section 16. Independent generation loses quality for theoretical rather than incidental reasons. The client-side model on which the design depends sits at the weak end of measured planning ability. Volunteer computing has been contracting for twenty years. And several measurements have gone against the design: the coherence criterion was not met, the ρ -tax curve and the confidence map's reliability claim remain withdrawn, M4 is falsified, and with a matched output budget fragmenting **costs** in all five families measured.

What survives is specific and is enough to continue. The $\perp$ curve — the axis this version makes central — has now been measured with a genuine fragmented arm, and it measures the design's thesis directly: **extracting with the model and aggregating with code beats the model alone** (T08R3), and the whole-document control separates the two mechanisms (T08R4). At 80 rows the gain comes from taking the arithmetic away from the model and splitting adds nothing; at 160 rows single-call extraction collapses and only splitting sustains it. A small node extracts reliably up to some task size, and fragmenting is what keeps each task below that threshold. The scope is narrow — few documents, one family, two sizes — and §15.4 declares it as such. Add to that a control that failed as it was required to, a verified grading instrument, a harness that refuses when its instrument cannot decide, and a set of models that say in advance what would kill them. What remains to settle is where that threshold lies and how it moves with the node class, and whether a context budget exists that satisfies coherence, privacy, verifiability and worker capability at once, at a cost below the value of the aggregated capacity.

This document is published now for the same reason that previously held it back: **the models it proposes were falsifiable, and they have now faced their test**. The verdicts are inside — one falsified, one corrected, one standing, two unmeasured — and Section 15.6 says how to attack the remaining ones and what each costs. With this publication, elements E19-E24 become prior art.

If it works, the result is not a cheaper way to buy what is already sold. It is inference capacity that grows with the number of people who participate rather than with the amount of capital available to build, held under a licence and a governance structure designed so that no single party can enclose it. That is worth attempting even at a substantial probability of failure, and it is worth attempting in the open, where it can be checked.

19. References

Numbering. Markers [1]–[90] preserve version 1.4's numbering, so that a citation from that document still points at the same source. Markers [92]–[135] are new in this version and correspond to the bibliographic review of the fragmentation fundamentals; all of them were verified against the online source indicated, and where only the abstract could be confirmed this is noted. Entries that carried forward an unconfirmed identifier from v1.4 were completed against the source for this publication. Those still marked $\triangle$ are only partly confirmed and say so, rather than being filled in from memory.

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